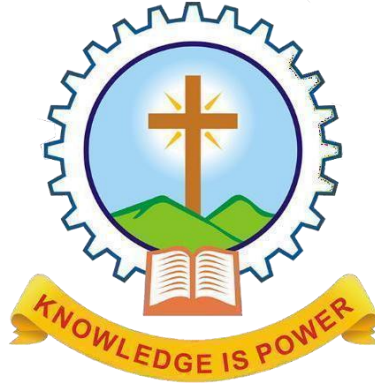


**MAR ATHANASIOUS COLLEGE OF ENGINEERING
KOTHAMANGALAM**

(Government Aided Autonomous Institution)



Department of Computer Applications

Mini Project Report

**START UP FAILURE RISK PREDICTOR USING
MACHINE LEARNING**

Done by

SINJU MATHEWS

Reg No: MAC24MCA-2051

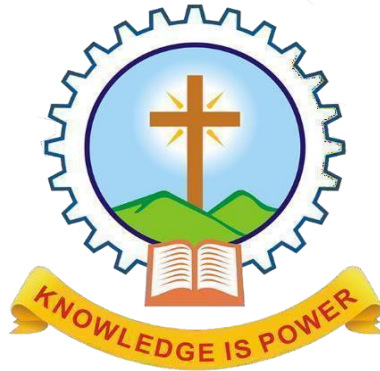
Under the guidance of

Dr. Sonia Abraham

2024-2026

**MAR ATHANASIOUS COLLEGE OF ENGINEERING
KOTHAMANGALAM**

**(Government Aided Autonomous Institution)
CERTIFICATE**



**Start Up Failure Risk Predictor using
Machine Learning**

Certified that this is the Bonafide record of project work done by

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during the third semester, in partial fulfilment of requirements for
award of the degree,

**Master of Computer Applications
of**

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ACKNOWLEDGEMENT

With heartfelt gratitude, I extend my deepest thanks to the Almighty for His unwavering grace and blessings that have made this journey possible. May His guidance continue to illuminate my path in the years ahead.

I am immensely thankful to Prof. Biju Skaria, Head of the Department of Computer Applications, Dr. Sonia Abraham, my mini project guide, Prof. Manu John and Prof. Minu Maria Joseph, our dedicated project coordinators, for their invaluable guidance and timely advice, which played a pivotal role in shaping this project. Their guidance, constant supervision, and provision of essential information were instrumental in the successful completion of the mini project.

I extend my profound thanks to all the professors in the department and the entire staff at MACE for their unwavering support and inspiration throughout my academic journey. My sincere appreciation goes to my beloved parents, whose guidance has been a beacon in every step of my path.

I am also grateful to my friends and individuals who generously shared their expertise and assistance, contributing significantly to the fulfillment of this endeavor.

ABSTRACT

Start-up failure is a persistent challenge in the entrepreneurial landscape, with a large proportion of new ventures unable to survive beyond their initial stages. Early identification of failure risk can significantly enhance strategic planning for founders, guide investors toward better decisions, and reduce the financial and operational losses associated with unsuccessful ventures. Previous studies have demonstrated the potential of machine learning in this domain, with models such as Random Forest achieving accuracies of up to 82%. Research further highlights that integrating multi-dimensional features spanning Investor, Business, and Market factors and employing ensemble algorithms like XGBoost can greatly improve predictive power, particularly on large datasets such as Crunchbase.

Building on this foundation, the present project develops a robust machine learning system to classify start-up outcomes into two categories: Success (0) or Failure (1). Using the publicly available “Crunchbase Startup Dataset” containing around 66,368 records, the study performs extensive preprocessing, data cleaning, and feature engineering to extract meaningful predictive patterns. Two advanced algorithms, Random Forest and XGBoost, are evaluated, with XGBoost emerging as the superior approach. The model achieved an accuracy of 91.03% and a ROC-AUC score of 0.9626, demonstrating strong discriminatory ability. These results not only align with existing literature but also further establish XGBoost as a powerful and reliable tool for data-driven assessment of start-up viability, offering valuable insights for researchers, entrepreneurs, and investors alike.

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1. INTRODUCTION

Start-ups are a vital engine of economic growth, driving innovation and generating employment across the globe. However, the start-up ecosystem is highly uncertain, with many ventures failing to survive their initial years due to challenges such as lack of funding, poor market alignment, and intense competition. These failures lead to substantial losses of capital, time, and potential societal benefits. As a result, accurately predicting a start-up's likelihood of success or failure has become an important focus for investors and founders, who increasingly rely on data-driven insights to guide strategic decisions.

Machine learning has emerged as a powerful tool for analyzing large-scale business data and uncovering patterns that traditional evaluation methods often miss. In this project, we develop a machine learning-based prediction model using the publicly available Crunchbase Startup Dataset to classify start-ups as either Success or Failure. The dataset includes diverse features such as funding history, company age, and geographic location, providing a strong foundation for predictive modeling.

To identify the most effective approach, we compare two ensemble algorithms Random Forest and XGBoost after performing extensive preprocessing and feature engineering. Both models have been widely recognized in previous research for their strong performance on structured datasets, particularly those involving entrepreneurial and financial data. Studies using Crunchbase have reported accuracies between 79–82%, highlighting the effectiveness of ensemble methods and the value of integrating investor, business, and market-related features.

Our findings show that XGBoost outperforms Random Forest across key evaluation metrics and is therefore selected as the final predictive model. Beyond prediction, the project aims to identify the most influential factors contributing to start-up failure, providing meaningful insights for stakeholders. In the future, this model can be deployed as a user-friendly web application to assist in real-time risk assessment, contributing to more informed and sustainable decision-making within the start-up ecosystem.

2. SUPPORTING LITERATURE

2.1 Literature Review

Paper 1 – Bidgoli MR, Vanani IR, Goodarzi M. Predicting the success of startups using a machine learning approach. *J Innov Entrep*. 2024;13(1):1–27.

This research addresses one of the most critical challenges in the entrepreneurial ecosystem — the inability of many early-stage startups to survive and grow due to high levels of uncertainty in funding, market adoption, and operational sustainability. The study focuses on developing a predictive analytics framework that can assist investors in identifying potentially successful startups before committing resources.

The dataset was collected from Crunchbase and LinkedIn, covering a 7-year period (March 2014 – March 2021). Features such as LinkedIn followers, employee count, Twitter followers, and last funding amount were chosen as indirect indicators of startup growth, visibility, and funding capability. Data preprocessing involved cleaning, handling missing values, encoding categorical variables, and feature scaling.

Five classification algorithms: Random Forest (RF), Gradient Boost (GB), Multilayer Perceptron (MLP), Logistic Regression (LR), and Support Vector Machine (SVM) — were applied. RF and GB outperformed the others with accuracies of 82% and 80%, respectively. SHAP values and permutation importance identified LinkedIn presence and funding amount as the most significant predictors.

Clustering was also performed using K-means, Hierarchical Clustering, and DBSCAN. K-means achieved the highest silhouette score (72%), indicating well-separated clusters. The combination of classification and clustering enabled predictive performance and actionable startup segmentation. The paper concludes with policy recommendations tailored for different startup clusters, enhancing investor decision-making.

TITLE	Predicting the Success of Startups Using a Machine Learning Approach
AREA OF WORK	Focuses on developing an integrated predictive and descriptive model for early-stage startup evaluation. Uses both classification to estimate success probability and clustering to group startups for targeted investment strategies.
DATASET	Crunchbase and LinkedIn datasets (2014–2021) containing features such as LinkedIn followers, Twitter followers, employee count, and last funding amount, serving as proxies for growth, visibility, and funding capability.
METHODOLOGY	Includes data preprocessing (cleaning, handling missing values, encoding, scaling), classification using RF, GB, MLP, LR, and SVM, feature importance via SHAP and permutation analysis, and clustering with K-means, Hierarchical, and DBSCAN. Policy recommendations are then developed for each cluster.
ALGORITHMS	Classification: Random Forest (RF), Gradient Boost (GB), Multilayer Perceptron (MLP), Logistic Regression (LR), Support Vector Machine (SVM). Clustering: K-means, Hierarchical Clustering, DBSCAN.
RESULTS	Random Forest achieved the highest accuracy (82%), followed by Gradient Boost (80%). K-means delivered the best clustering performance with a silhouette score of 72%.
ADVANTAGES	Combines predictive accuracy with descriptive insights; interpretable through SHAP values; offers actionable policies for investors.
FUTURE PROPOSAL	Extend to real-time data integration, adapt to sector-specific startup ecosystems, and incorporate dynamic clustering that responds to changing market conditions.

Table 2.1 Literature Review Summary-I

Paper 2 – Gangwani D, Zhu X, Furht B. Exploring investor–business–market interplay for business success prediction. *J Big Data*. 2023;10(1):1–28.

This study introduces the Investor–Business–Market (IBM) interplay framework, which integrates investor profiles, business fundamentals, and market conditions to improve startup success prediction. The authors argue that traditional models relying solely on business attributes, such as sector or founding year, overlook the significant role played by investor capacity and external market forces.

The dataset includes companies ranging from early-stage startups to Fortune 1000 firms, enabling the analysis of businesses at different growth stages. Success is defined strictly as IPO or acquisition, while failure is identified as closure. To reduce ambiguity, operating firms were excluded since their final outcomes remain unknown.

Feature engineering was performed across three domains: Investor features (experience, funding history, geographic scope), Business features (founding year, industry, innovation measures), and Market features (sector growth rate, competitive intensity, regional density). This multi-dimensional approach allowed the models to capture the dynamic interplay between stakeholders.

Multiple machine learning algorithms were applied, with Random Forest and XGBoost delivering the strongest performance. Models built using the IBM framework achieved significantly higher accuracy and AUC scores compared to single-dimension models, validating the importance of a holistic perspective.

A key contribution of this work lies in its alignment with real-world investment practices, where investors consider not only business strength but also who is backing the company and the environment in which it operates. The authors conclude that the IBM framework provides more trustworthy, interpretable, and actionable predictions, making it a valuable advancement in the field of startup success modeling.

TITLE	Exploring Investor–Business–Market Interplay for Business Success Prediction
AREA OF WORK	Proposes the IBM framework to combine three dimensions of startup evaluation — Investor, Business, and Market — to create a holistic and realistic predictive model for business success.
DATASET	Covers companies from startups to Fortune 1000 firms; success defined as IPO or acquisition; failure as closure. Operating companies excluded to avoid bias. Features represent investor history, business sector, and market growth.
METHODOLOGY	Define success/failure; engineer IBM features; train models using RF and XGBoost; compare performance of multi-dimensional models against single-domain models; assess accuracy and AUC improvements.
ALGORITHMS	Random Forest XGBoost
RESULTS	IBM-based models significantly outperform single-domain models in predictive accuracy and AUC, showing the value of integrating investor and market data with business metrics.
ADVANTAGES	Reduces bias by excluding ambiguous data; interpretable through grouped features; aligns closely with real investor evaluation processes.
FUTURE PROPOSAL	Adapt IBM framework to specific industries, integrate macroeconomic and sectoral data, and develop dynamic scoring systems that update with market trends.

Table 2.2 Literature Review Summary-II

Paper 3 – Gautam L, Wattanapongsakorn N. Machine learning models to investigate startup success in venture capital using Crunchbase dataset. In: Proceedings of the 21st International Joint Conference on Computer Science and Software Engineering (JCSSE); 2024; IEEE. p. 475–81.

This paper addresses the growing reliance of venture capital (VC) firms on data-driven methods to manage the overwhelming number of startup proposals they receive annually, often exceeding several thousand. Traditional manual evaluations are not only time-intensive but also subject to human bias, making it difficult to systematically identify high-potential startups. To overcome these challenges, the authors propose the application of advanced machine learning (ML) models to build a predictive framework that improves decision-making in VC deal flow.

The study utilizes the Crunchbase enterprise dataset, one of the most comprehensive sources of startup and business information, covering more than 2.2 million companies worldwide. The dataset includes variables such as founding year, location, industry classification, number of funding rounds, and funding amounts. These attributes were carefully preprocessed through cleaning, missing value imputation, and normalization. In addition, feature selection techniques were employed to retain only the most influential predictors of startup performance, thereby reducing noise and improving model interpretability.

Several ML algorithms were evaluated, with a focus on XGBoost, LightGBM, and Logistic Regression. Performance was measured using accuracy and F1-score, ensuring that both precision and recall were considered. Results demonstrated that boosting algorithms outperformed traditional models, with XGBoost and LightGBM achieving 79–80% accuracy and an F1-score of 86%, while Logistic Regression lagged behind.

A key insight from the paper is that machine learning should act as a decision-support tool rather than a replacement for human expertise. The proposed models are envisioned as a filtering mechanism, allowing VC firms to efficiently prioritize promising startups for further in-depth analysis. The authors recommend integrating such ML systems into deal-flow management pipelines, enabling dynamic startup scoring that can be continuously updated with real-time funding, market, and social signals.

TITLE	Machine Learning Models to Investigate Startup Success in Venture Capital Using Crunchbase Dataset
AREA OF WORK	Aims to improve venture capital efficiency by predicting startup success likelihood using one of the largest available datasets, helping prioritize proposals for deeper review.
DATASET	Crunchbase dataset (~2.2 million companies) with variables including funding history, founding year, location, and industry classification.
METHODOLOGY	Data preprocessing and cleaning; feature selection to identify relevant predictors; model training with XGBoost, LightGBM, and Logistic Regression; evaluation based on accuracy and F1-score.
ALGORITHMS	XGBoost LightGBM Logistic Regression
RESULTS	Boosting models (XGBoost, LightGBM) achieved 79–80% accuracy and 86% F1-score, outperforming Logistic Regression..
ADVANTAGES	Extremely large dataset; high predictive accuracy; suitable for integration into VC deal-flow management systems.
FUTURE PROPOSAL	Test with international datasets, integrate real-time funding trends, and embed models into interactive decision-support dashboards.

Table 2.3 Literature Review Summary-I

2.2 Literature Review Summary

TITLE	SUMMARY
Bidgoli et al. Bidgoli MR, Vanani IR, Goodarzi M. Predicting the success of startups using a machine learning approach. J Innov Entrep. 2024;13(1):1–27.	The paper develops a hybrid ML framework using classification (RF, GB, MLP, LR, SVM) and clustering (K-means, Hierarchical, DBSCAN) on Crunchbase–LinkedIn data (2014–2021). RF achieved 82% accuracy and K-means 72% silhouette score. The study highlights interpretability via SHAP and provides cluster-based investor policy insights. Future work suggests real-time and sector-specific extensions
Gangwani et al. Gangwani D, Zhu X, Furht B. Exploring investor–business–market interplay for business success prediction. J Big Data. 2023;10(1):1–28.	This paper introduces the IBM framework combining Investor, Business, and Market features. Using data from startups to Fortune 1000 firms, success was defined as IPO/acquisition and failure as closure. RF and XGBoost showed that IBM models outperform single-dimension ones, improving accuracy and AUC. The framework aligns with investor practices; future work suggests domain-specific and dynamic scoring models.
Gautam & Wattanapongsakorn Gautam L, Wattanapongsakorn N. Machine learning models to investigate startup success in venture capital using Crunchbase dataset. In: Proceedings of the 21st International Joint Conference on Computer Science and Software Engineering (JCSSE); 2024; IEEE. p. 475–81.	The study applies ML to VC decision support using Crunchbase (~2.2M companies). Features include sector, funding, and location. Models tested: XGBoost, LightGBM, Logistic Regression. Boosting methods achieved 79–80% accuracy and 86% F1, outperforming LR. Authors propose integrating ML into VC deal-flow systems with real-time market data for dynamic evaluation.

Table 2.4 Overall Summary of all papers

2.3 Findings and Proposals

The collective insights from the three reviewed studies confirm that machine learning is highly effective in predicting startup success and failure, with ensemble algorithms such as Random Forest, Gradient Boosting, XGBoost, and LightGBM consistently achieving strong results. These models utilize diverse features ranging from funding history and company age to investor profiles and market conditions, capturing both internal and external drivers of success.

From Bidgoli et al. (2024), the key insight is that combining classification with clustering improves both predictive accuracy and interpretability. Their Random Forest model achieved 82% accuracy, while clustering revealed meaningful groupings of startups that could guide investor-specific strategies. This demonstrated that prediction frameworks should not stop at accuracy metrics but also provide actionable insights for decision-making.

From Gangwani et al. (2023), the major contribution is the Investor–Business–Market (IBM) framework, which incorporates investor capacity and market dynamics along with business fundamentals. Their results showed that multi-domain models outperform those relying only on company attributes, underlining the importance of external ecosystem features such as competition density and sector growth when modeling startup outcomes.

From Gautam & Wattanapongsakorn (2024), the critical finding is that boosting algorithms such as XGBoost and LightGBM excel on large-scale datasets, achieving close to 80% accuracy and an F1-score of 86% using Crunchbase data. Their emphasis on ML as a decision-support tool, rather than a replacement for human expertise, offers a practical roadmap for integrating predictive models into venture capital workflows.

Taken together, these findings highlight three priorities: leveraging diverse features, using robust ensemble models, and ensuring interpretability for real-world use. Building on these lessons, the proposed work will compare Random Forest and XGBoost using the Crunchbase dataset. The best-performing model will be deployed in a web-based dashboard, enabling venture capitalists and policymakers to make data-driven, timely investment decisions while improving the early identification of high-risk startups.

3. SYSTEM ANALYSIS

3.1 Analysis of the dataset

3.1.1. About the dataset

The Startup Dataset is a comprehensive and valuable resource for analyzing business characteristics and predicting startup success or failure. It contains a mix of financial, categorical, and temporal features that enable exploration of factors influencing startup performance and long-term survival.

Dataset Source: Crunchbase-based Startup Dataset.

<https://www.kaggle.com/datasets/yanmaksi/big-startup-secses-fail-dataset-from-crunchbase>

Size: The dataset includes information on 66,368 startups, with attributes capturing funding history, geographic details, business categories, and important lifecycle milestones such as founding date and funding rounds.

Data Structure: It contains multiple features such as `funding_total_usd`, `funding_rounds`, `founded_at`, `first_funding_at`, `last_funding_at`, `category_list`, and `country_code`, along with the target variable `status`. These features support a variety of analyses, ranging from financial performance trends and industry-based comparisons to predictive modeling of startup outcomes.

Sparsity: The dataset contains missing values in certain attributes, particularly in `homepage_url`, `category_list`, and geographic details (`country_code`, `state_code`, `region`, `city`). Despite these gaps, the dataset remains highly informative, providing sufficient coverage to perform correlation analysis, feature engineering, and machine learning–based prediction of startup survival versus failure.

3.1.2. Explore the Dataset

A) Statistical Analysis

a) Dataset Overview

The dataset consists of 66,368 records and 14 columns, focusing on startup identity, funding details, geographic attributes, and lifecycle events.

- Total Records: 66,368
- Total Features: 13 independent variables + 1 target variable (status)
- Data Types: Mixture of categorical (e.g., category_list, country_code) and numerical (e.g., funding_total_usd, funding_rounds) features.
- Missing Values: Present in homepage_url, category_list, and location-related attributes.

Feature Categories:

1. Identifiers and Descriptors – permalink, name, homepage_url
2. Business Characteristics – category_list, funding_total_usd, funding_rounds
3. Geographic Information – country_code, state_code, region, city
4. Temporal Features – founded_at, first_funding_at, last_funding_at
5. Target Variable – status (operating, acquired, ipo, closed → later mapped to Success = 0, Failure = 1)

```
# Shape of dataset (rows, columns)
print("Shape:", df.shape)
print("Data Types:", df.dtypes)
```



```
Shape: (66368, 14)
Data Types: permalink      object
name                      object
homepage_url              object
category_list              object
funding_total_usd          object
status                    object
country_code               object
state_code                 object
region                     object
city                      object
funding_rounds             int64
founded_at                 object
first_funding_at           object
last_funding_at            object
dtype: object
```

Fig 3.1 shape and datatype of dataset

The startup dataset is dominated by categorical attributes, with only a limited set of numerical features. Among them, `funding_rounds` is a clean integer field, while `funding_total_usd` is stored as an object and requires preprocessing to convert it into a numeric format before detailed analysis.

b) Dataset Sample Records

The first few records of the dataset show the structure and variety of information available. This helps provide an initial understanding of how startups are described in terms of funding, geography, and lifecycle status.

```
pd.set_option('display.max_colwidth', 15)
# First 5 rows
df.head()
```

index	permalink	name	homepage_url	category_list	funding_total_usd	status	country_code
0	/organization/-fame	#fame	http://livfame.com	Media	10000000	operating	IND
1	/organization/-qounter	:Qounter	http://www.qounter.com	Application Platforms Real Time Social Network Media	700000	operating	USA
2	/organization/-the-one-of-them-inc-	(THE) ONE of THEM, Inc.	http://oneofthem.jp	Apps Games Mobile	3406878	operating	NaN
3	/organization/0-6-com	0-6.com	http://www.0-6.com	Curated Web	2000000	operating	CHN
4	/organization/004-technologies	004 Technologies	http://004gmbh.de/en/004-interact	Software	-	operating	USA

Fig 3.2 First Five rows of dataset

1 to 5 of 5 entries  

country_code	state_code	region	city	funding_rounds	founded_at	first_funding_at	last_funding_at
IND	16	Mumbai	Mumbai	1	NaN	2015-01-05	2015-01-05
USA	DE	DE - Other	Delaware City	2	2014-09-04	2014-03-01	2014-10-14
NaN	NaN	NaN	NaN	1	NaN	2014-01-30	2014-01-30
CHN	22	Beijing	Beijing	1	2007-01-01	2008-03-19	2008-03-19
USA	IL	Springfield, Illinois	Champaign	1	2010-01-01	2014-07-24	2014-07-24

Fig 3.3 First Five rows of dataset continuation

c) Descriptive Statistics (Numerical Features)

Descriptive statistics (`df.describe()`) revealed that funding amounts and funding rounds are highly skewed, with most startups having very little funding and only a few raising very large amounts.

funding_rounds	
count	66368.000000
mean	1.732522
std	1.360251
min	1.000000
25%	1.000000
50%	1.000000
75%	2.000000
max	19.000000

Fig 3.4 Description of dataset

d) Missing Values Analysis

Missing values analysis highlighted that categorical and location-based features have higher missingness, requiring imputation strategies.

df.isnull().sum()	
	0
permalink	0
name	1
homepage_url	5058
category_list	3148
funding_total_usd	0
status	0
country_code	6958

state_code	8547
region	8030
city	8028
funding_rounds	0
founded_at	15221
first_funding_at	24
last_funding_at	0
dtype: int64	

Fig 3.5 List of missing values in dataset

e) Target Variable Distribution (Raw Table)

The raw target variable distribution (status) showed multiple categories such as *operating*, *acquired*, *ipo*, and *closed*.

```
# Raw status distribution (before mapping to success/failure)
raw_status = df['status'].value_counts(dropna=False)
print(raw_status)

status
operating    53034
closed       6238
acquired     5549
ipo          1547
Name: count, dtype: int64
```

Fig 3.6 Raw target variable distribution in dataset

B) Data Visualization Analysis

a) Status Distribution

The bar chart shows that far more startups are still operating compared to those that have succeeded or failed. This highlights how few startups reach exit events like IPO/acquisition and how many continue without formally shutting down.

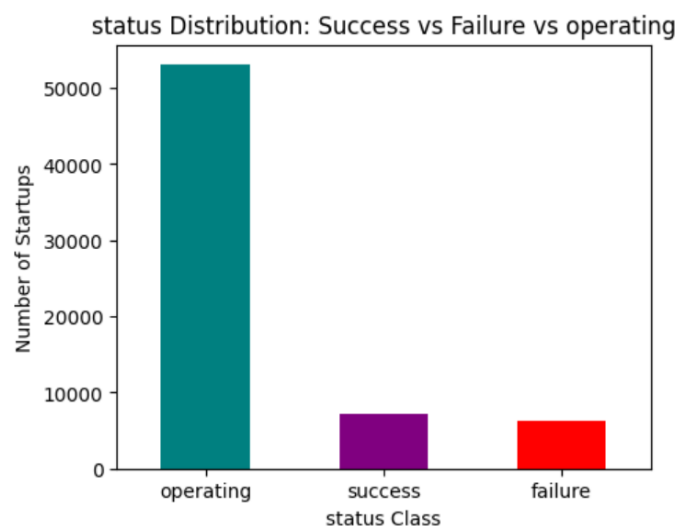


Fig 3.7 Status distribution plot

b) Funding Distribution

Most startups received very minimal funding, while only a tiny fraction secured extremely large investments. The log-scale plot reveals clearer patterns by reducing the effect of billion-dollar outliers.

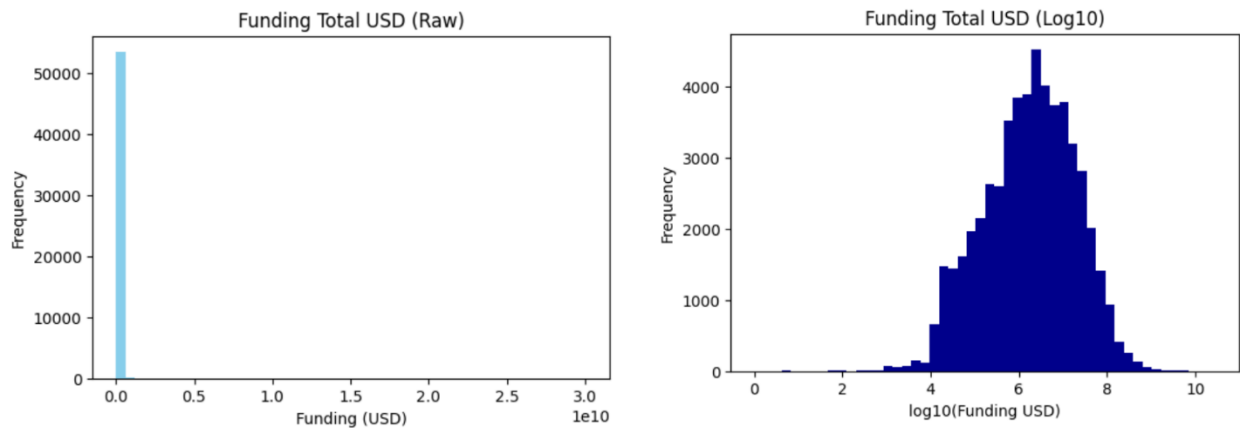


Fig 3.8 Funding distribution

c) Funding Rounds Distribution

Most startups raise only 1–3 funding rounds, with higher rounds becoming increasingly rare. Startups that achieve more rounds show stronger long-term survival, suggesting consistent investor confidence. The distribution highlights how competitive and selective later-stage funding becomes.

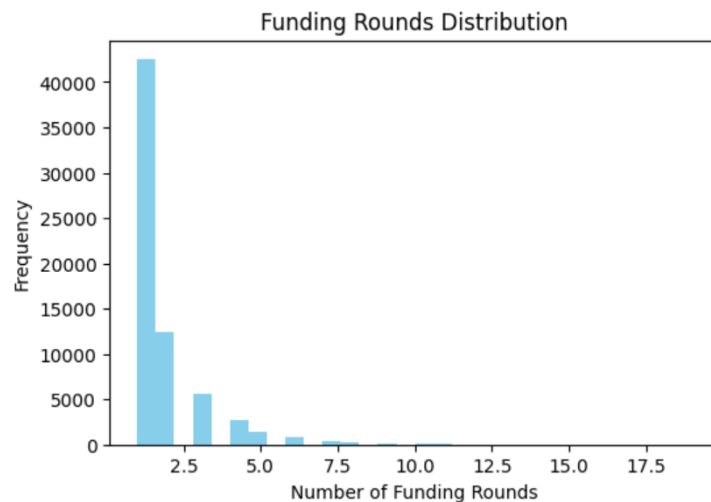


Fig 3.9 Funding rounds distribution

d) Top Startup Categories

The category-wise chart shows that industries like Biotechnology, Software, E-commerce, and Advertising dominate the startup landscape. These sectors attract more founders due to large markets and scalability. Their higher representation indicates ongoing global demand and investment focus in these fields.

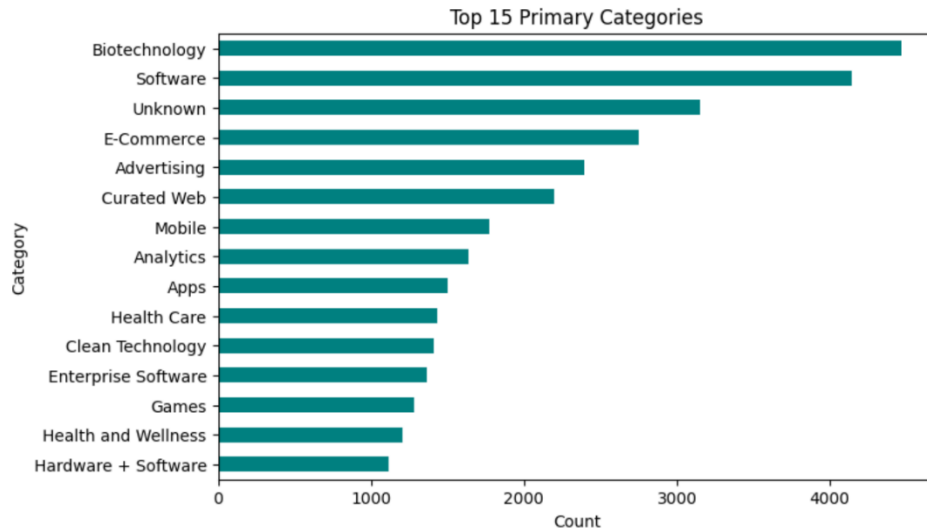


Fig 3.10 To startup Categories

e) Geographic Distribution

The geographic distribution reveals that the USA leads by a wide margin in the number of startups. It is followed by countries like the UK, India, Canada, and China, which also have strong entrepreneurial ecosystems. This pattern reflects differences in funding availability, infrastructure, and innovation culture.

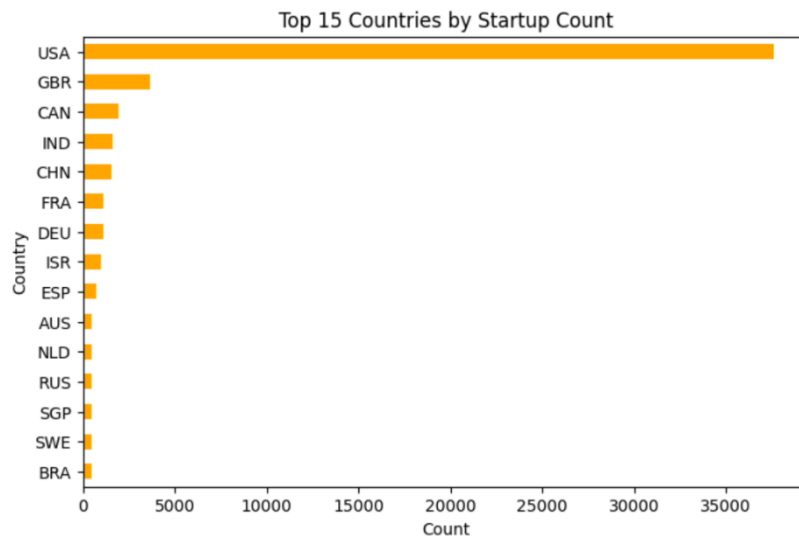


Fig 3.11 Geographic Distribution

f) Year-wise Startup Formation Trend

The yearly trend chart shows a major rise in startup formations between 1998 and 2012. This period aligns with the dot-com boom, the growth of internet technologies, and expanding venture capital activity. The surge reflects a global wave of digital innovation during those years.

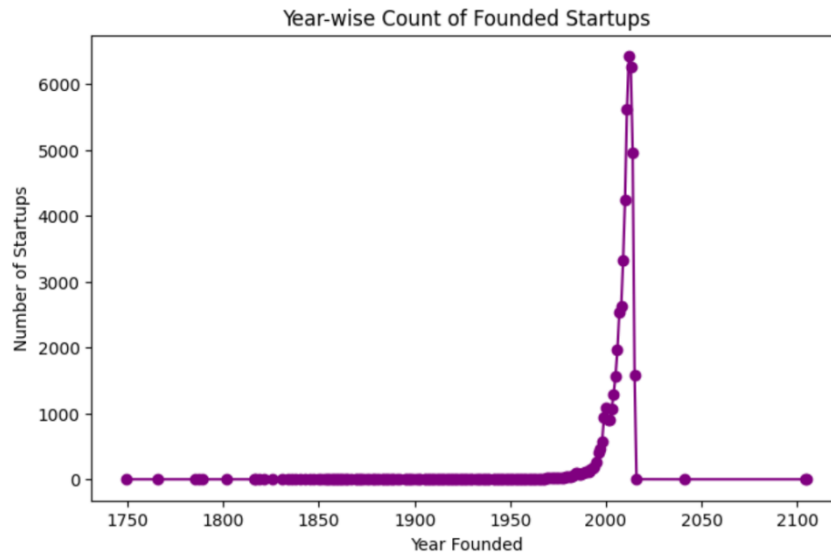


Fig 3.12 Year-wise startup trends

C) Insights and Conclusions of EDA

The exploratory analysis of the dataset revealed several important patterns that provide a deeper understanding of startup dynamics. First, funding emerged as a strong differentiator between successful and failed startups. Startups that survived or achieved acquisition/IPO status typically secured significantly higher amounts of funding and participated in multiple funding rounds, while failed startups often remained underfunded or limited to just one or two rounds. This suggests that access to capital and investor confidence are critical to long-term survival.

Another key insight relates to company longevity. Startups that have been in existence for longer periods, measured by the difference between their founding date and most recent funding event, demonstrate higher survival rates. This indicates that time in the market contributes to stability and resilience, with older companies being more likely to withstand the volatility of the startup ecosystem.

The analysis also highlighted the importance of geographic location. Startups based in the USA dominate the dataset not only in sheer numbers but also in success rates, reflecting the strength

of established ecosystems such as Silicon Valley and New York. These regions provide access to better funding, mentorship, and infrastructure, offering startups a significant survival advantage.

At the industry level, certain categories, particularly Biotechnology and Software, show consistently higher probabilities of success compared to others such as Advertising or Consumer Products. This could be attributed to the higher investor interest, innovation potential, and scalability associated with these domains.

Finally, the data characteristics revealed that both funding amounts and funding rounds follow highly skewed distributions, with the majority of startups clustered at the lower end and a small number achieving disproportionately high values. This imbalance underscores the necessity of applying log-transformations, normalization, and careful feature scaling during preprocessing to avoid distortion in model training.

Collectively, these insights not only enrich the understanding of startup performance factors but also provide a strong foundation for feature engineering and predictive modeling. They highlight the critical role of funding, longevity, geography, and industry focus in shaping the likelihood of startup success or failure.

3.2 Data Preprocessing

3.2.1 Data Cleaning

Data cleaning was a critical step to ensure the dataset was consistent, reliable, and ready for further analysis. The raw dataset contained several missing values, inconsistencies in numeric fields, and unformatted dates. The following operations were performed:

a) Numeric Conversion and Dataset Optimization

- Converted 'funding_total_usd' to numerics.
- Filled missing funding values with 0 (means no funding reported)
- Dropped unnecessary columns (including 'permalink', 'homepage_url', 'region', 'city', 'name')

```

After Cleaning
(66368, 9)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 66368 entries, 0 to 66367
Data columns (total 9 columns):
#   Column              Non-Null Count  Dtype
---  ---
0   category_list        63220 non-null  object
1   funding_total_usd    66368 non-null  float64
2   status               66368 non-null  object
3   country_code         59410 non-null  object
4   state_code           57821 non-null  object
5   funding_rounds       66368 non-null  int64
6   founded_at           51147 non-null  object
7   first_funding_at     66344 non-null  object
8   last_funding_at      66368 non-null  object
dtypes: float64(1), int64(1), object(7)
memory usage: 4.6+ MB

```

Fig 3.13 Numeric Conversion and Dataset Optimization

b) Date conversion

Converted `founded_at`, `first_funding_at`, `last_funding_at` to datetime format.

```

--- Data Types After Cleaning ---
<class 'pandas.core.frame.DataFrame'>
Index: 51137 entries, 1 to 66366
Data columns (total 10 columns):
#   Column              Non-Null Count  Dtype
---  ---
0   category_list        49702 non-null  object
1   funding_total_usd    51137 non-null  float64
2   status               51137 non-null  object
3   country_code         51137 non-null  object
4   state_code           46327 non-null  object
5   funding_rounds       51137 non-null  int64
6   founded_at           51137 non-null  datetime64[ns]
7   first_funding_at     51114 non-null  datetime64[ns]
8   last_funding_at      51137 non-null  datetime64[ns]
9   funding_total_usd_num 41883 non-null  float64
dtypes: datetime64[ns](3), float64(2), int64(1), object(4)

```

Fig 3.14 Date conversion

c) Missing Values

- Imputed missing `country_code` with 'UNK'.
- Drop rows where `founded_at` or `last_funding_at` is missing (critical for timeline analysis).

```

--- Null Values After Cleaning ---
category_list      1435
funding_total_usd    0
status             0
country_code       0
state_code        4810
funding_rounds     0
founded_at         0
first_funding_at   23
last_funding_at    0
funding_total_usd_num 9254
dtype: int64

```

Fig 3.15 Missing Values

d) Outliers and Skewness

funding_total_usd_num was highly skewed with extreme values. We applied a log transform to normalize distribution and reduce impact of outliers.

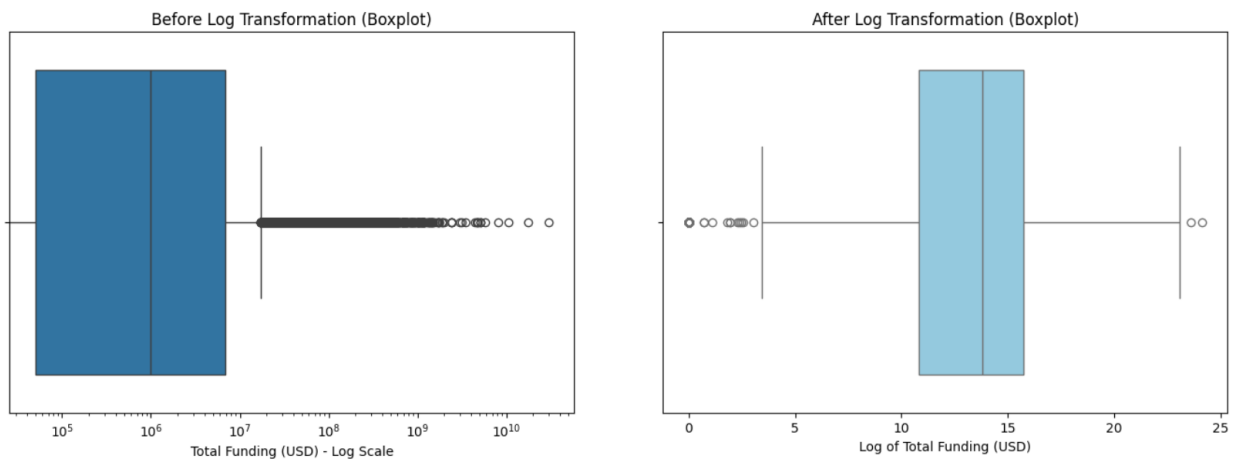


Fig 3.16 Finding outliers and skewing of funding

e) Derived Features

We engineered new features to capture company lifecycle and funding behavior:

- age_of_company_years = years between founding and last funding
- funding_duration_years = time span between first and last funding

```

--- Summary of Engineered Features ---
      company_age_years  funding_duration_years  funding_total_usd \
count      51114.000000      51114.000000      5.111400e+04
mean         7.244708         1.065062         1.593250e+07
std         10.285513         1.965242         1.870680e+08
min          0.000000         0.000000         0.000000e+00
25%          2.000000         0.000000         5.000000e+04
50%          5.000000         0.000000         1.000000e+06
75%          9.000000         1.451061         6.947283e+06
max        266.000000        29.587953         3.007950e+10

      funding_total_usd_log
count      51114.000000
mean         11.763002
std          5.945142
min          0.000000
25%         10.819798
50%         13.815512
75%         15.753861
max         24.127110

```

Fig 3.17 Features Engineered

f) Training labels Encoding

The target column status had multiple categories. We grouped them into a binary target:

- acquired, ipo : 0 (success)
- closed : 1 (failure)
- operating : NaN (unlabeled, will be predicted later)

```

Original status counts:
status
operating    41519
acquired     4332
closed       4076
ipo          1187
Name: count, dtype: int64

Training labels (0=success, 1=failure, NaN=operating):
model_label
NaN         41519
0.0         5519
1.0         4076
Name: count, dtype: int64

```

Fig 3.18 Training labels encoding

g) Encoding Categorical Features

Categorical variables were encoded for ML:

One Hot Encoding → creates a new binary (0/1) column for each unique category in a feature, indicating the presence of that category for each record.

For high-cardinality columns like country, rare categories can be grouped into “Other” before encoding to reduce dimensionality.

country_code_ARE	country_code_ARG	country_code_AUS	country_code_AUT	country_code_BEL	...	primary_category_finance
False	False	False	False	False	...	False
False	False	False	False	False	...	False
False	False	False	False	False	...	False
False	False	False	False	False	...	False
False	False	False	False	False	...	False

Fig 3.20 Encoded categorical variables

h) Handling Inconsistencies

Verifying dataset (labelled) quality by checking for:

- Duplicate records
- Negative funding values
- Logical contradictions (e.g., zero funding rounds but non-zero funding amount)

```
Final missing values after cleaning:
funding_rounds          0
funding_total_usd_log    0
company_age_years       0
funding_duration_years   0
model_label             0
..
primary_category_security 0
primary_category_semiconductors 0
primary_category_software 0
primary_category_unknown  0
primary_category_web hosting 0
Length: 124, dtype: int64
Duplicate rows before removal: 500
Duplicate rows after removal: 0
New shape: (9095, 124)
Invalid rows (zero rounds but nonzero funding): 0
```

Fig 3.19 Inconsistencies verified

i) Status Classification (Operating)

Classified startups with the status “*operating*” into “*success*”(0) or “*failure*”(1) categories using the trained model. These predicted labels are then added to form the complete dataset used for final model training and evaluation.

```
Operating startups classified:
predicted_label
1      23008
0      18511
Name: count, dtype: int64
```

Fig 3.21 operating Classification

j) Combined Final Labels

Combined already labelled data with classified operating datapoints to get the final dataset with proper distribution.

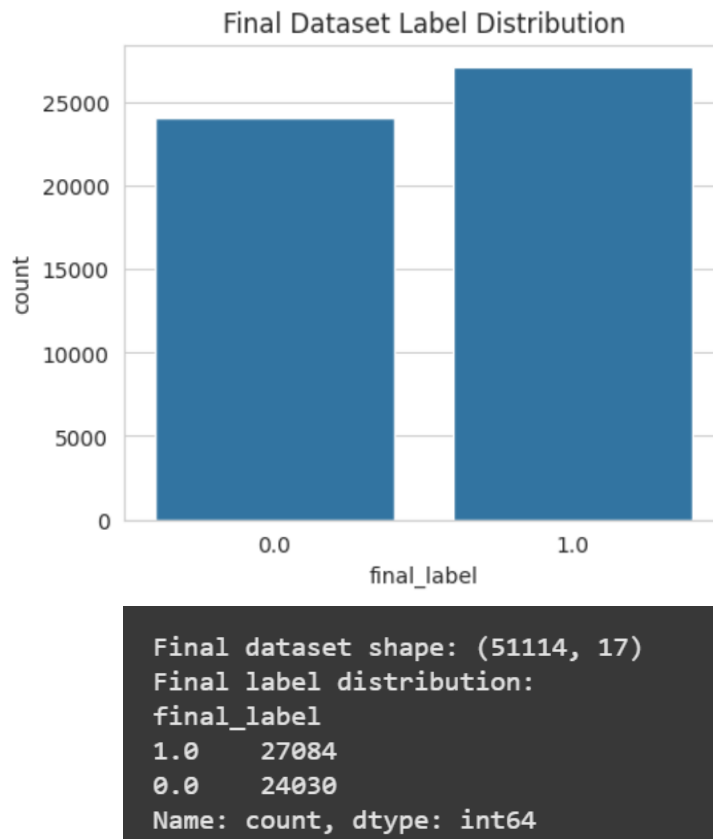


Fig 3.22 Final dataset distribution

3.2.2 Analysis of Feature Variables

Feature extraction is the process of identifying which attributes (columns) in the dataset are most relevant for predicting the target variable (here: success/failure of startups). Not all features contribute equally — some may be redundant, weak, or even misleading. By applying statistical and model-based methods, we can prioritize the strongest predictors and reduce noise, which leads to better and faster model training.

A) Correlation Analysis

Measures the strength of linear relationships between numerical features and the target variable. Helps detect redundant features (highly correlated with each other) or weak features (low correlation with the target).

Interpretation:

Correlation close to +1 / -1 → strong relationship.

Correlation close to 0 → weak or no linear relationship.

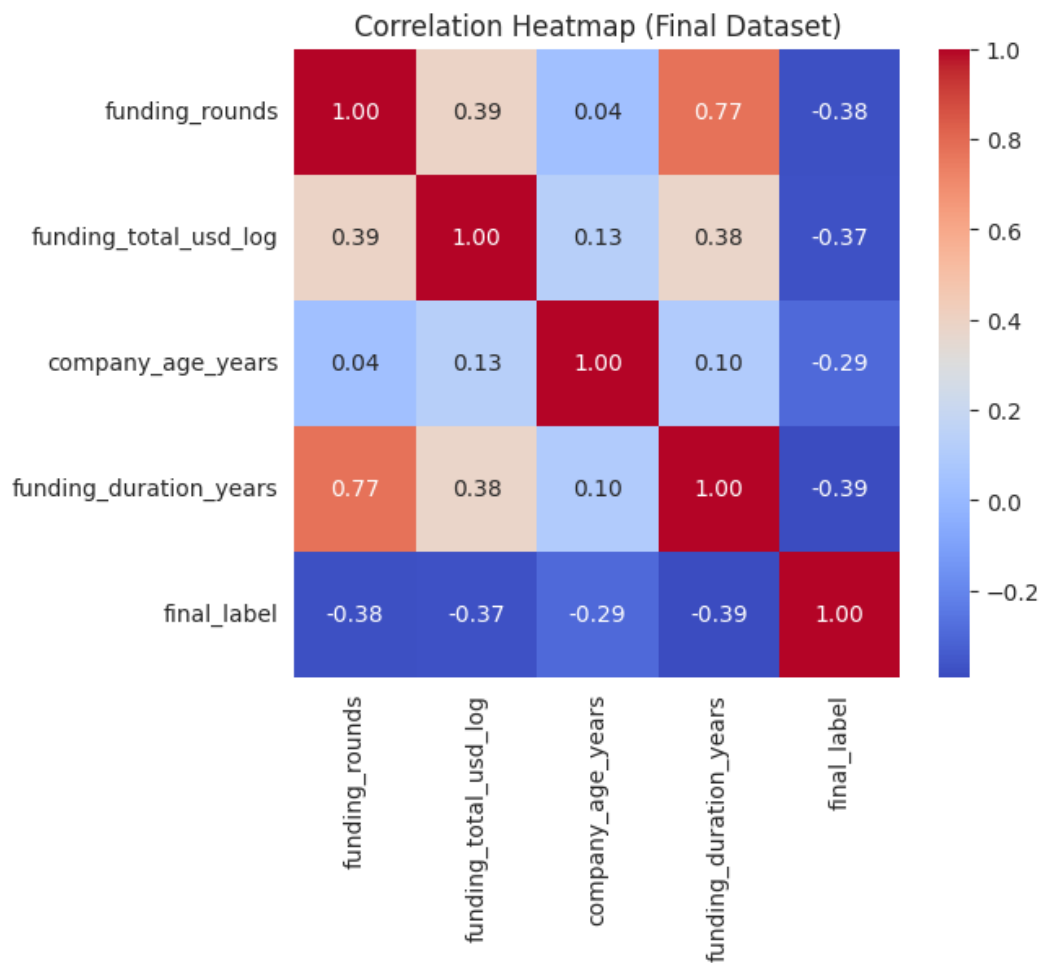


Fig 3.23 Correlation analysis heatmap

- Funding rounds and funding duration years show a strong positive correlation (0.77) startups with longer funding periods tend to have undergone more funding rounds.
- Funding amount (log) is moderately correlated with both funding rounds (0.39) and funding duration (0.38) more funding rounds and longer durations are usually associated with higher total funding.

- Company age has weak correlations with other features, suggesting it contributes independent information about maturity.
- Final label (success/failure) shows moderate negative correlations with funding-related metrics (around -0.37 to -0.39) indicating that startups with fewer rounds, lower funding, or shorter funding durations are more likely to fail.

Overall takeaway: Funding rounds, funding amount, and funding duration are the most influential features for predicting startup outcomes, while company age provides complementary but weaker predictive value.

B) Pre-processed Dataset

a. Shape and info

```
Shape of the final encoded dataset ready for training: (51114, 174)

Info of the final encoded dataset for training:
<class 'pandas.core.frame.DataFrame'>
Index: 51114 entries, 15 to 66366
Columns: 174 entries, funding_rounds to primary_category_web hosting
dtypes: bool(169), float64(3), int32(1), int64(1)
memory usage: 10.4 MB
```

Fig 3.24 Shape and description of pre-processed dataset

Gives a quick summary of the dataset, including the number of rows, columns, data types, non-missing values, and memory usage.

After preprocessing, the dataset contains 174 columns, of which 169 are boolean indicator variables derived through one-hot encoding of categorical attributes.

The following three categorical features were encoded:

- `primary_category` – the industry or domain in which the startup operates (e.g., Software, Biotechnology, E-commerce).
- `country_code` – the country in which the startup is headquartered (e.g., USA, GBR, IND).

- `state_code` – the state or regional administrative code associated with the startup's location (e.g., CA, NY, BER).

In addition, the dataset includes four numerical features:

- `funding_rounds` – total number of funding rounds the startup has completed.
- `funding_total_usd_log` – log-transformed value of the total funding amount raised (in USD).
- `company_age_years` – the age of the company at the time of data collection, measured in years.
- `funding_duration_years` – the duration, in years, between the startup's first and most recent funding rounds.

Finally, the dataset contains the binary classification label:

- `final_label` : 0 → success, 1 → failure

b. First five records of the dataset

First 5 rows of the final dataset:

	<code>funding_rounds</code>	<code>funding_total_usd_log</code>	<code>company_age_years</code>	<code>funding_duration_years</code>	<code>final_label</code>	<code>country_code_ALB</code>	<code>country_code_ARE</code>
15	1	15.424949	3	0.000000	0.0	False	False
20	1	13.122365	6	0.000000	0.0	False	False
23	2	14.745705	5	1.125257	0.0	False	False
31	2	14.038655	4	0.076660	0.0	False	False
32	1	17.370859	15	0.000000	0.0	False	False

5 rows × 174 columns

Fig 3.25 First five records of preprocessed dataset

c. Last five records of the dataset

Last 5 rows of the final dataset:

	<code>funding_rounds</code>	<code>funding_total_usd_log</code>	<code>company_age_years</code>	<code>funding_duration_years</code>	<code>final_label</code>	<code>country_code_ALB</code>	<code>country_code_ARE</code>
66361	1	14.803787	8	0.000000	1.0	False	False
66362	3	13.928840	1	0.925394	0.0	False	False
66364	4	11.646626	3	2.329911	0.0	False	False
66365	1	0.000000	4	0.000000	1.0	False	False
66366	1	9.808792	1	0.000000	1.0	False	False

5 rows × 174 columns

Fig 3.26 Last five records of preprocessed dataset

d. Statistics summary

Statistical summary of selected features:					
	funding_rounds	funding_total_usd_log	company_age_years	funding_duration_years	final_label
count	51114.000000	51114.000000	51114.000000	51114.000000	51114.000000
mean	1.846324	11.763002	7.244708	1.065062	0.529874
std	1.454868	5.945142	10.285513	1.965242	0.499112
min	1.000000	0.000000	0.000000	0.000000	0.000000
25%	1.000000	10.819798	2.000000	0.000000	0.000000
50%	1.000000	13.815512	5.000000	0.000000	1.000000
75%	2.000000	15.753861	9.000000	1.451061	1.000000
max	19.000000	24.127110	266.000000	29.587953	1.000000

Fig 3.27 Statistics summary of preprocessed dataset

- **Sample size:** The dataset contains 51,114 startups after preprocessing.
- **Funding rounds:**
Average = 1.85, indicating most startups undergo 1–2 funding rounds.
Maximum = 19, suggesting a few highly funded startups with multiple investment stages.
- **Funding amount (log scale):**
Mean = 11.76, Median = 13.82, showing a right-skewed distribution — most startups receive lower funding, while a few have exceptionally high amounts.
Minimum = 0 implies startups with no or undisclosed funding.
- **Company age (years):**
Mean = 7.24 years, but the maximum age = 266 years indicates presence of some very old firms, possibly due to data anomalies or legacy companies.
50% of startups are ≤ 5 years old, confirming the dataset mainly represents young ventures.
- **Funding duration (years):**
Median = 0, meaning most startups receive funding quickly or only once.
However, the maximum of ~29.6 years shows a few long-running funding histories.
- **Final label (success/failure):**
Mean ≈ 0.53 , indicating a balanced dataset — around 53% successful and 47% failed startups.

C) Dataset Split

To prepare our data for machine learning, we first perform a crucial step: splitting it into training and testing sets. This ensures we can train our model and then test its performance on data it has never seen before.

- Training Set (70%): The model uses this data to learn the underlying patterns and relationships between a startup's features and its success or failure.
- Test Set (30%): This data is kept separate and is used only at the end to evaluate how well the trained model generalizes to new, unseen examples.

We use `train_test_split()` with `stratify=y_final` to ensure that the proportion of successful and failed startups is the same in both the training and test sets as it is in the original dataset. Finally, numerical features are standardized using `StandardScaler` to bring them onto a common scale.

```
# --- 1. Train/test split (Unbalanced version) ---
X_train, X_test, y_train, y_test = train_test_split(
    X_final, y_final, test_size=0.3, random_state=42, stratify=y_final
)

# Scale (unbalanced)
scaler_unbalanced = StandardScaler()
scaler_unbalanced.fit(X_train[numerical_features])

X_train_unbal = X_train.copy()
X_test_unbal = X_test.copy()

Train: (35779, 174) | Test: (15335, 174)
```

Fig 3.28 Dataset Split

3.2.3 Analysis of Class Variables

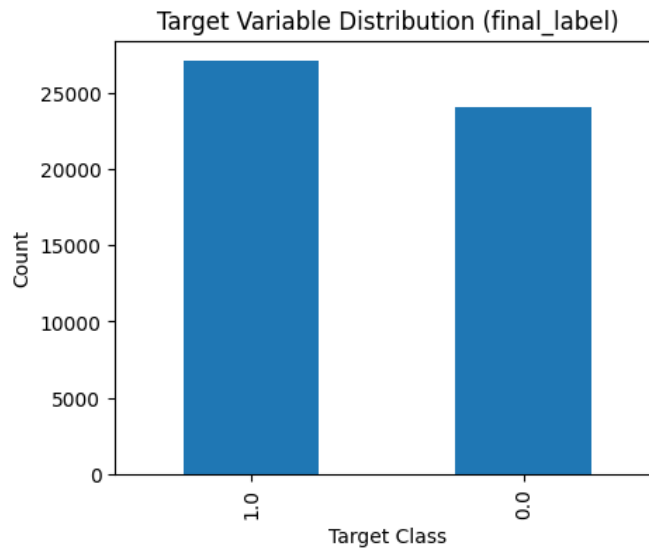


Fig 3.29 Target Class Distribution

The bar chart compares the number of:

1 → Failed startups

0 → Successful startups

Key Insight :

- The dataset is fairly balanced:
27,500 : unsuccessful
24,000 : successful
- This is good for modeling because it reduces bias toward one class.

A relatively balanced dataset helps machine learning models learn decision boundaries more effectively.

3.3 Data Visualization

a) Funding Rounds vs Target

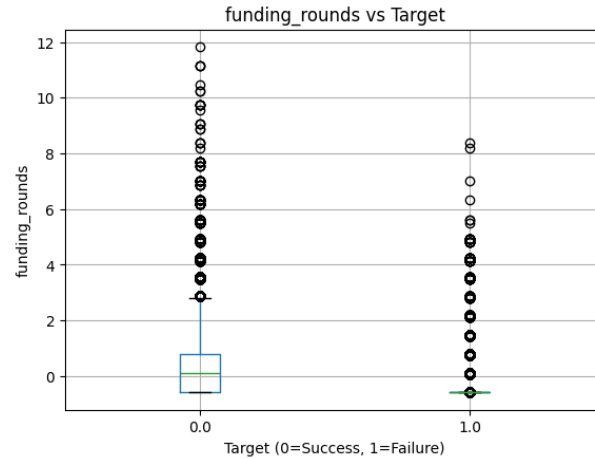


Fig 3.30 Funding Rounds vs Target

Startups that succeed usually go through more funding rounds, while failed ones have fewer. More rounds signal repeated investor confidence, making `funding_rounds` a strong positive indicator of success.

b) Total Funding (Log) vs Target

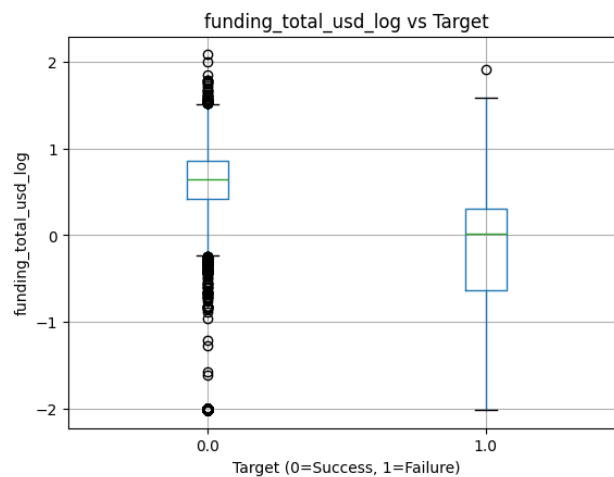


Fig 3.31 Total Funding vs Target

Successful startups have much higher total funding compared to failed ones. Higher funding improves growth and survival, making `funding_total_usd_log` one of the most influential predictors.

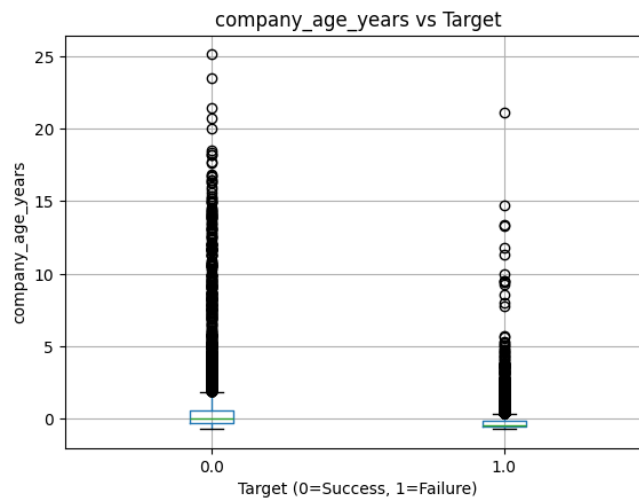
c) Company Age vs Target

Fig 3.32 Company Age vs Target

Both successful and failed startups show similar age distributions. Age doesn't meaningfully separate outcomes, so `company_age_years` contributes limited predictive power.

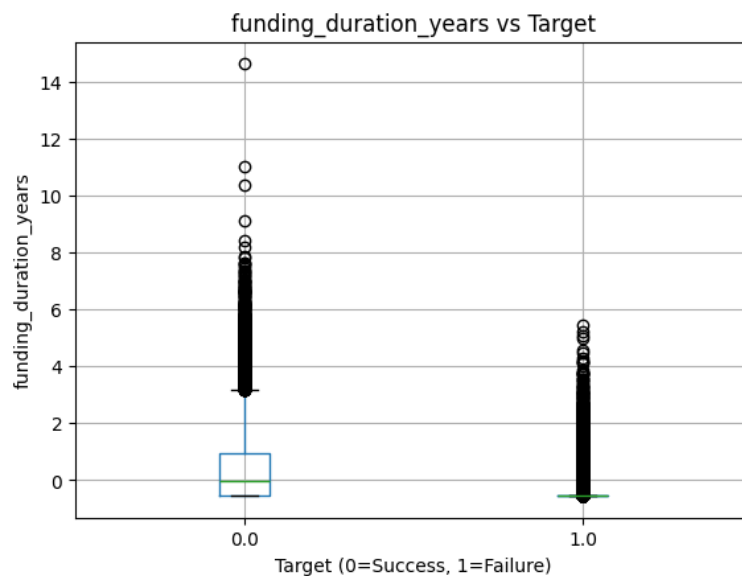
d) Funding Duration vs Target

Fig 3.33 Funding Duration vs Target

Failed startups typically take longer to secure funding. Longer fundraising suggests difficulty attracting investors, making `funding_duration_years` a strong risk indicator.

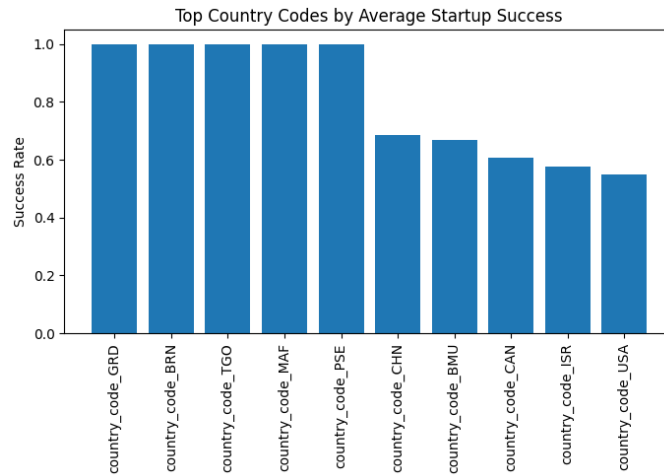
e) Country Success Rates

Fig 3.34 Country success Rate

Success rates vary widely across countries. Differences in ecosystem maturity, funding availability, and market conditions make `country_code` an important categorical predictor.

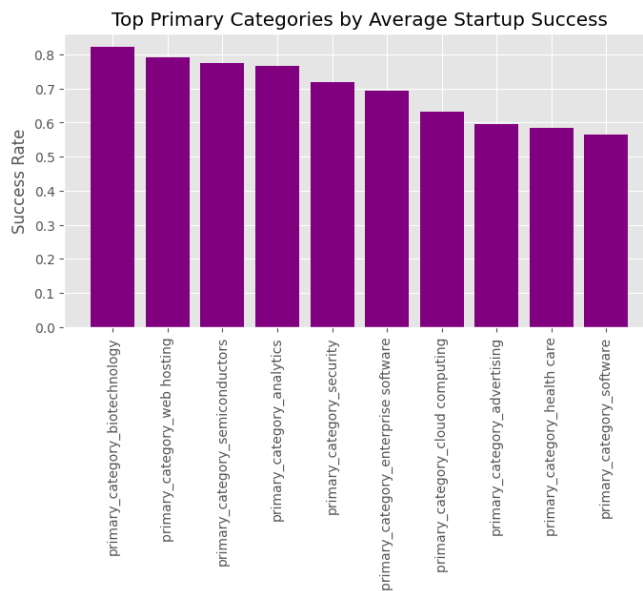
f) Primary Category Success Rates

Fig 3.35 primary Category Success Rate

Some sectors especially tech show much higher success rates. Categories with stronger demand and scalability make `primary_category` a valuable predictive feature.

3.4 Analysis of Architecture Algorithm

3.4.1 Proposed Algorithms

i) Random Forest (RF)

a. Description & Justification

Algorithm Random Forest is an ensemble method that builds multiple decision trees on random subsets of data and features, then averages results (majority voting for classification).

Why chosen:

- Handles non-linear relationships well.
- Resistant to overfitting due to bagging technique.
- Provides feature importance rankings, useful in identifying most influential startup characteristics.

b. Flowchart

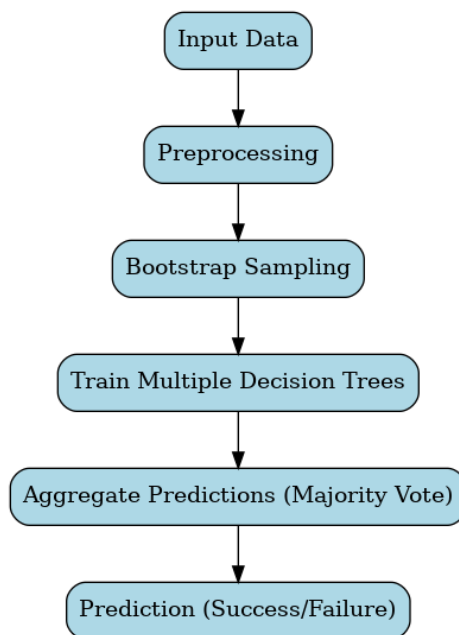


Fig 3.36 Flow chart of Random Forest

c. Pseudocode

Input: Training dataset (X, Y)

Step 1: For each tree in the forest:

- Take a random sample of data (with replacement)
- Select random subset of features

- Grow a decision tree using chosen data & features

Step 2: Repeat until N trees are trained

Step 3: For prediction:

- Pass input through all trees
- Collect predictions
- Final output = Majority vote (classification)

ii) XGBoost (Extreme Gradient Boosting)

a. Description & Justification

XGBoost is an advanced gradient boosting algorithm that builds decision trees sequentially, where each new tree corrects errors made by previous ones.

Why chosen:

- Highly accurate and efficient, especially with structured tabular data like startups dataset
- Handles missing values, imbalanced classes, and large datasets effectively.
- Known to outperform many other models in Kaggle competitions.

b. Flowchart

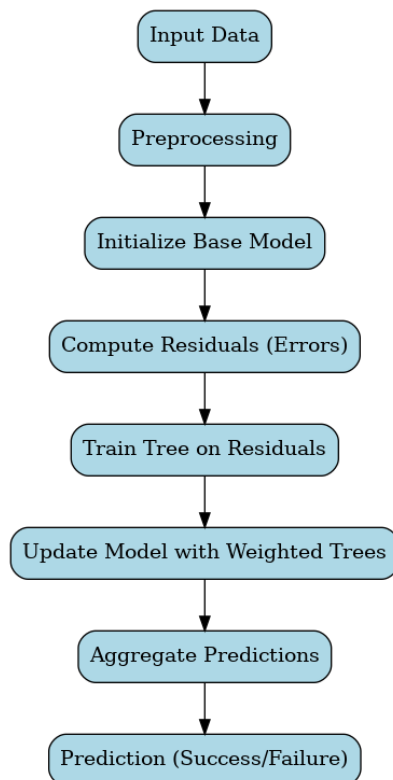


Fig 3.37 Flow chart of XGBoost

c. Pseudocode

Input: Training dataset (X, Y)

Step 1: Initialize predictions with base value (e.g., average)

Step 2: For each boosting round:

- Compute residuals (errors of current model)
- Train a decision tree on residuals
- Compute leaf weights to minimize loss
- Update model by adding weighted tree

Step 3: For prediction:

- Sum predictions from all trees (weighted)
- Apply sigmoid function
- Threshold probability → Final classification

Objective of the project

The objective of this project is to predict startup failure risk (binary classification: *success* or *failure*) using historical, financial, and demographic features of startups. Two machine learning algorithms Random Forest (RF), and Extreme Gradient Boosting (XGBoost) are implemented for performance comparison to identify the most effective model.

- Random Forest (RF) is selected for its ability to capture non-linear feature interactions, robustness against overfitting, and feature importance ranking.
- XGBoost is included because of its high predictive power, efficiency with large datasets, and strong handling of complex relationships, making it one of the top-performing algorithms in structured data problems.

These algorithms are trained and tested on the Crunchbase Startup Dataset, their performances are compared, and the best-performing algorithm is chosen based on accuracy and other binary classification evaluation metrics.

3.5 Project Pipeline

3.5.1 Project pipeline diagram

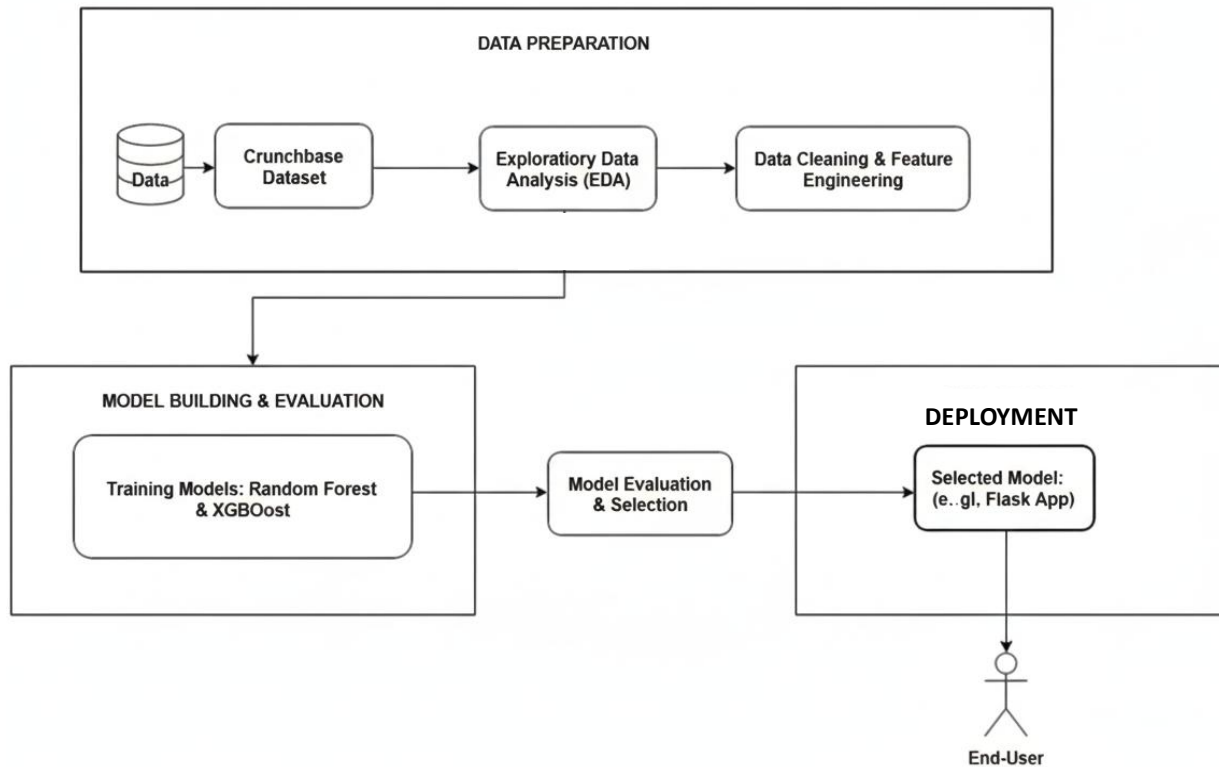


Fig 3.38 Project workflow

- **Data Preparation**

The process begins with the Crunchbase Startup Dataset, which includes historical data on start-up characteristics such as funding amounts, key dates, industry categories, and geographic locations. Exploratory Data Analysis (EDA) is then performed to gain insights into the dataset. EDA helps visualize distributions, identify correlations between features and outcomes, detect outliers, and uncover meaningful patterns that guide the subsequent steps. Since the raw data contains noise and inconsistencies, the next step is data cleaning and feature engineering. This involves handling missing values, encoding categorical variables like `country_code` and `category_list`, and creating new, impactful features such as 'Company Age' and 'Time to First Funding' to better capture the venture's lifecycle.

- **Model Building**

After data preparation, the cleaned and engineered dataset is used for training the machine learning models. In this project, two powerful ensemble algorithms are applied: Random Forest and XGBoost. Random Forest is known for its robustness and ability to handle complex datasets by combining multiple decision trees, which reduces the risk of overfitting. XGBoost, an advanced gradient boosting method, is chosen for its high predictive accuracy and computational efficiency. Each model is trained on the same data and then evaluated using key performance metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. This comparison allows for identifying the strengths and weaknesses of each algorithm, and based on this evaluation, the best-performing model is selected for deployment.

- **Deployment**

Once the best model is identified, it is deployed to make it accessible for practical use. Model deployment involves integrating the chosen model into a system, such as a web application or an API, where new data can be input for real-time predictions. This step bridges the gap between research and real-world application, allowing the model to be actively used for decision-making. In the context of start-up failure prediction, the deployed model would enable an end-user, such as an investor or founder, to input a start-up's characteristics and receive an instant prediction about its likelihood of success or failure. The model's output provides valuable, data-driven insights that can be used for risk assessment and strategic planning.

3.6 Feasibility Analysis

A feasibility study examines the strengths, weaknesses, resources, and overall viability of a proposed system. For smartphone addiction prediction system project, the feasibility has been evaluated in the following categories:

- Technical Feasibility
- Economic Feasibility
- Operational Feasibility

3.6.1 Technical Feasibility

Technical feasibility examines whether the technology required for this project is available and whether the development team possesses the expertise to implement it effectively.

Data Processing and Analysis

Feasibility:

The project requires efficient handling and analysis of behavioural, demographic, and usage-related data. Using pandas and NumPy ensures reliable preprocessing capabilities such as handling missing values, detecting inconsistencies, encoding features, and preparing the dataset for machine learning models. Visualization is made feasible using Matplotlib and Seaborn, which support a wide range of graphical analyses.

Expertise:

The team has demonstrated strong familiarity with pandas, NumPy, Matplotlib, and Seaborn for cleaning, feature extraction, trend visualization, and exploratory data analysis. This confirms the technical capability to manage and analyze raw data efficiently.

Machine Learning Algorithms

Feasibility:

The project employs Random Forest and XGBoost, two powerful, industry-standard machine learning algorithms widely used for prediction tasks involving structured behavioural data. Both algorithms are well-supported and suitable for classification tasks related to smartphone addiction prediction, ensuring high technical feasibility.

Expertise:

The successful implementation of Random Forest and XGBoost—along with hyperparameter tuning, model evaluation, accuracy comparison, and feature importance analysis—demonstrates strong competence in machine learning methodologies and optimization techniques.

Data Visualization

Feasibility:

Visual analysis of behavioural trends, class distribution, and key features is conducted using Matplotlib and Seaborn, both of which are highly reliable and widely adopted visualization libraries. These tools fully support the project's visualization requirements.

Expertise:

The generated plots—including histograms, box plots, heatmaps, and distribution charts—show clear proficiency in visual analytics, helping interpret patterns in smartphone usage and identify key predictive indicators.

File Handling

Feasibility:

Storing, reading, and managing datasets is handled using **pandas**, which makes it technically feasible to preprocess and save datasets in appropriate formats such as CSV. This ensures smooth integration with machine learning workflows and deployment modules.

Expertise:

The proper handling of data cleaning, encoding, feature generation, and storing processed datasets indicates strong technical ability in dataset management and preparation for model training.

3.6.2 Economic Feasibility

Economic feasibility evaluates whether the system is financially viable in terms of development and long-term usage.

Feasibility:

The system uses open-source technologies such as Python, Flask, pandas, NumPy, Random Forest, XGBoost, and visualization libraries, keeping software costs minimal. Development requires only basic hardware resources, making the overall cost highly economical.

Benefits:

Early detection of smartphone addiction can promote healthier digital habits, making the project valuable from a social and wellness perspective. The benefits greatly outweigh the minimal development costs.

Maintenance Costs**Feasibility:**

Maintenance costs remain low as updates primarily include dataset refresh, occasional retraining of Random Forest and XGBoost models, and minor updates to the Flask interface.

Benefits:

As predictions become more accurate with fresh data, long-term user benefit and system performance continue to improve with minimal recurring expenses.

3.6.3 Operational Feasibility

Operational feasibility assesses whether the system meets user needs and can be smoothly adopted in practical use.

User Acceptance**Feasibility:**

The system is designed to provide users with clear, data-driven insights into their smartphone usage and addiction tendencies. This transparency improves acceptance and encourages users to adopt healthier digital behaviors.

Integration**Feasibility:**

The predictive models are integrated into a lightweight Flask web application, allowing users to enter usage information and receive predictions instantly. This smooth integration enhances operational efficiency and ease of implementation.

Scalability

Feasibility:

As user volume increases, computational load on Random Forest and XGBoost may also increase due to larger datasets.

Mitigation:

Scalability can be maintained through:

- Incremental model updating
- Optimized retraining
- Using cloud-based computing (optional)

This ensures the system remains responsive even with growing data.

Ease of Use

Feasibility:

A simple UI developed using HTML and CSS ensures ease of use. Clearly labelled input fields allow users to enter their smartphone usage details effortlessly. Predictions are shown in a clean, readable format requiring no technical knowledge.

Training:

Minimal to no user training is required, as the system is designed to be intuitive and user-friendly.

The software environment includes all tools and platforms required for development, testing, and deployment.

3.7 System Environment

3.7.1 Software Environment

Programming Languages

- **Python:** Primary programming language due to its simplicity and availability of powerful ML and data-processing libraries.

Data Processing and Analysis

- pandas & NumPy: For cleaning, transforming, and analyzing behavioural data.
- Matplotlib & Seaborn: For generating visual insights and understanding dataset patterns.

Machine Learning Libraries

- Random Forest & XGBoost: Core algorithms used for addiction prediction.
- scikit-learn: Supports preprocessing, model evaluation, encoding, feature scaling, and train-test splitting.

Development Environment

- Google Colab: Cloud-based environment enabling efficient data handling, model training, and experimentation.
- Visual Studio Code: Used for organizing and writing Python scripts, Flask code, and project structuring.

Web Development

- HTML & CSS: Used to design and structure the user interface for the Flask application.

Web Framework

- Flask: Lightweight Python web framework used to deploy machine learning models and provide a prediction interface.

Version Control

- GitHub: For code backup, version management, and project updates.

3.7.2 Hardware Environment

The hardware environment specifies all the physical resources required to support development and execution.

Computing Resources

- CPU: Intel Core i5
- Processor Memory (RAM): 16 GB
- Cache: 12 MB
- Storage: 512 GB SSD
- Monitor: 14.6" Display

- Keyboard: Standard 108-key Enhanced Keyboard
- Mouse: Optical Mouse

These specifications provide more than sufficient performance for running data preprocessing workflows, model training, visualizations, and Flask-based deployment without lag.

4. SYSTEM DESIGN

4.1 Models Building

The model building phase of the start-up failure prediction project establishes a structured methodology to ensure the accuracy, reliability, and practical utility of the final classifier. This stage defines the project objectives, selects appropriate high-performance algorithms, and outlines the modeling approach for effective risk prediction.

1. **Objective:**

Develop a machine learning system to predict the binary outcome of a start-up—Success (0) or Failure (1). The prediction is based on key historical features such as funding data, company timeline, and geographical location.

2. **Approach:**

Implement and compare two powerful supervised machine learning algorithms Random Forest (RF) and XGBoost to identify the best-performing model for the task.

- Random Forest: Chosen for its robustness, ability to handle complex and high-dimensional data, and its inherent resistance to overfitting through ensemble learning.
- XGBoost: Selected for its state-of-the-art performance, speed, and high predictive accuracy, making it a leading choice for classification tasks.

The final model will be selected based on a comprehensive evaluation using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC score.

3. **Data Preparation:**

- Import and preprocess the "Crunchbase Startup Dataset" from Kaggle.
- Handle missing values, perform data type conversions, and filter out irrelevant or inconsistent data points.
- Apply a log transformation to the `funding_total_usd` feature to normalize its highly skewed distribution.
- Engineer new features like `company_age_years` and `funding_duration_years` to capture temporal dynamics.

- Encode categorical features (country_code, primary_category, etc.) using one-hot encoding.
- Split the final dataset into training (70%) and testing (30%) subsets for model development and evaluation.

4. Exploratory Data Analysis (EDA):

- Examine distributions of key features like funding_total_usd and funding_rounds.
- Identify trends, such as the peak in start-up founding years, through visualizations like histograms and line charts.
- Analyze relationships between variables using a correlation heatmap to inform feature selection.
- Verify the balance of the target variable (Success vs. Failure) to guide the modeling strategy.

5. Model Training:

- Train the Random Forest and XGBoost classifiers on the preprocessed dataset.
- Evaluate each model's performance on the unseen test set.
- Compare the results to determine the most effective algorithm for start-up failure prediction.
- Save the best-performing model (xgboost_model.pkl) for deployment in the final application.

This structured model planning framework ensures a systematic approach to building an accurate and reliable start-up failure prediction system, combining advanced machine learning techniques with data-driven insights.

4.1.1 Implementation Code

The implementation integrates two powerful ensemble-based machine learning algorithms XGBoost and Random Forest to evaluate their ability to predict startup success or failure. The script begins by importing essential libraries for data handling, model training, visualization, and performance evaluation. The preprocessed training and testing datasets are loaded, after which the features (X) and target variable (y) are separated for both sets.

For XGBoost, the model is configured with optimized hyperparameters such as *n_estimators*, *max_depth*, *learning_rate*, and *subsample*, allowing it to learn complex nonlinear patterns effectively. The Random Forest classifier is also trained with balanced class weighting to

handle class imbalance in the dataset. After training, both models generate predictions on the training and testing sets, followed by performance evaluation using metrics such as accuracy, classification reports, confusion matrices, ROC curves, and ROC–AUC scores.

Visualizations including the confusion matrix, ROC curve, learning curve, and feature importance plots help assess model behavior, overfitting tendencies, and the influence of key predictors. Finally, the trained models are saved using joblib for later use, completing a comprehensive machine-learning pipeline from training to evaluation and storage.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.ensemble import RandomForestClassifier
import xgboost as xgb
from sklearn.metrics import classification_report, confusion_matrix
from sklearn.model_selection import learning_curve
from sklearn.metrics import ConfusionMatrixDisplay
import joblib
```

Fig 4.1 Importing libraries

<pre>X_train = train_df.drop('final_label', axis=1) y_train = train_df['final_label'] X_test = test_df.drop('final_label', axis=1) y_test = test_df['final_label'] # --- 2. Train Model --- rf_model = RandomForestClassifier(n_estimators=200, random_state=42, class_weight='balanced', n_jobs=-1) rf_model.fit(X_train, y_train)</pre>	<pre># --- 2. Train Model --- xgb_model = xgb.XGBClassifier(objective='binary:logistic', n_estimators=300, learning_rate=0.05, max_depth=6, subsample=0.8, colsample_bytree=0.8, eval_metric="logloss", random_state=42, base_score=0.5) xgb_model.fit(X_train, y_train)</pre>
--	--

Fig 4.2 Implementing algorithms

4.1.2 Model Planning

To ensure reliable prediction performance, the dataset was divided into training and testing subsets using a 70:30 ratio, with stratification applied to maintain the original class distribution. This prevents bias during model training, especially in the presence of an imbalanced target variable. The loading process involves reading preprocessed CSV files that already contain cleaned and engineered features suitable for model training.

Both XGBoost and Random Forest were chosen due to their robustness, ability to handle nonlinear relationships, and compatibility with high-dimensional tabular data. Hyperparameters for XGBoost were carefully set to balance model complexity and generalization, while Random Forest utilized class weighting to better handle minority-class samples. The planned workflow includes training the models, evaluating them using accuracy and other classification metrics, and generating additional diagnostics such as learning curves and feature importance rankings. This structured approach ensures the models are trained effectively, assessed thoroughly, and validated on unseen test data to measure real-world predictive performance.

```
# --- 1. Train/test split ---
X_train, X_test, y_train, y_test = train_test_split(
    X_final, y_final, test_size=0.3, random_state=42, stratify=y_final
)
```

Fig 4.2 Train-Test Split

4.1.3 Training

The training phase focuses on developing a predictive model for start-up failure using the supervised algorithms Random Forest (RF) and XGBoost. The preprocessed and feature-engineered dataset was used to train both models, ensuring they learned from a clean, structured, and informative set of data. Key features such as `funding_total_usd_log`, `company_age_years`, `funding_rounds`, and `funding_duration_years` were used to identify the financial and operational patterns linked to a start-up's outcome.

Each model was trained to achieve high predictive accuracy. While Random Forest offers strong performance by aggregating the results of many decision trees, XGBoost employs a more advanced gradient-boosting technique that sequentially builds models to correct the errors

of its predecessors, often leading to superior results. As part of the final data preparation, StandardScaler was used to scale the numerical features, ensuring that all variables contributed equally to the model's training process. The performance of both models was then evaluated on the test set using metrics like accuracy, precision, recall, and F1-score. Based on this comparison, the XGBoost model was selected as the best-performing model for deployment.

```

 RandomForest Model Evaluation:
 Training Accuracy: 0.9947
 Testing Accuracy: 0.9071

 XGBoost Model Evaluation:
 Training Accuracy: 0.9256
 Testing Accuracy: 0.9103

```

Fig 4.3 Trained Model Evaluation

4.1.4 Testing

The testing phase evaluates the performance of the startup failure prediction models and their ability to generalize to unseen data. Model evaluation is conducted using the testing portion of the dataset, which represents a diverse set of real-world startup profiles. Key performance metrics such as accuracy, precision, recall, and F1-score are calculated to assess each model's effectiveness in predicting startup outcomes. The final selection is based on which model demonstrates the highest reliability, particularly in identifying startups that are likely to fail.

```

 XGBoost Classification Report:
               precision    recall  f1-score   support

      0.0       0.8963       0.9151       0.9056        7209
      1.0       0.9233       0.9061       0.9146        8126

   accuracy          0.9103        15335
  macro avg          0.9106        15335
 weighted avg          0.9104        15335

 Random Forest Classification Report:
               precision    recall  f1-score   support

      0.0       0.8900       0.9157       0.9026        7209
      1.0       0.9232       0.8996       0.9112        8126

   accuracy          0.9071        15335
  macro avg          0.9066        15335
 weighted avg          0.9076        15335

```

Fig 4.4 Testing classification reports

5. RESULTS AND DISCUSSION

Confusion Matrix of each model

a) XGBoost

The XGBoost model achieved the superior performance for startup failure prediction, demonstrating the highest accuracy of approximately 91.0%. Critically, the model proved highly reliable in preventing the most costly errors, recording a low number of False Negatives (763) meaning very few failed startups were missed—and a low number of False Positives (612). This low error rate, especially for False Negatives, makes XGBoost the most robust choice for a system designed for risk assessment.

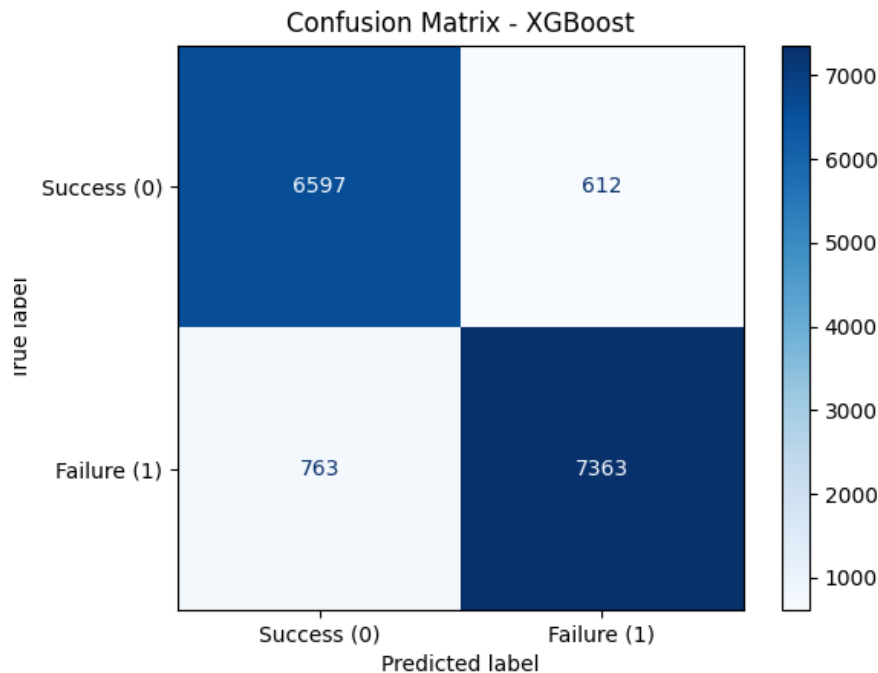


Fig 5.1 XGBoost confusion matrix

b) Random Forest

The Random Forest model emerged as the second-best classifier, showing good performance with an overall accuracy of 90.7%. While highly accurate, its confusion matrix revealed a slightly greater susceptibility to errors compared to XGBoost. It generated 816 False Negatives and 608 False Positives, indicating a slightly higher risk of missing true failure cases. This

performance, while respectable, positions it as a strong but less precise alternative to the XGBoost model.

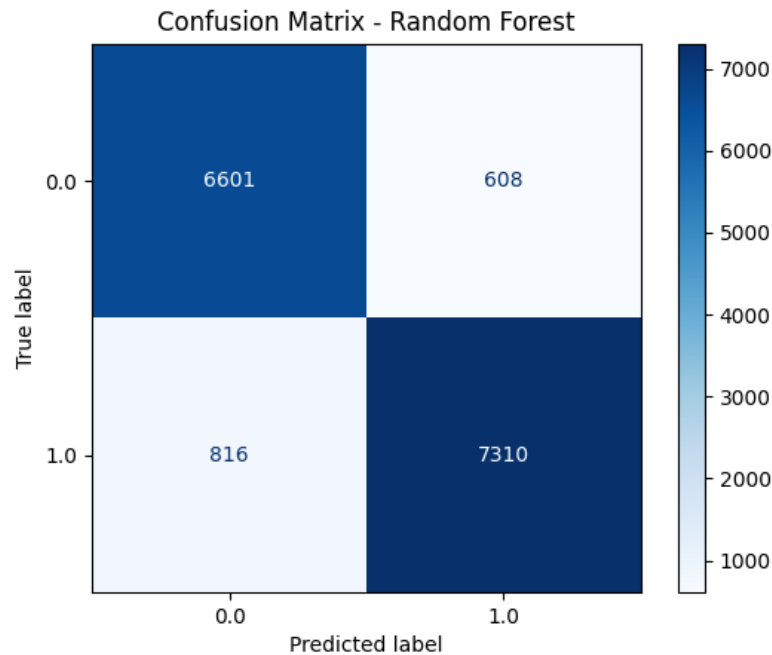


Fig 5.2 Random Forest confusion matrix

Accuracy and Classification Report

a) XGBoost

The XGBoost model performed exceptionally well, achieving a testing accuracy of 91.03%. Both success (0) and failure (1) classes were identified with high precision, recall, and F1-scores, all around 0.91. The low error metrics and strong balanced performance indicate that the model effectively distinguishes between outcomes, making it highly suitable for deployment.

```
XGBoost Classification Report:
              precision    recall  f1-score   support

     0.0       0.8963     0.9151     0.9056       7209
     1.0       0.9233     0.9061     0.9146       8126

 accuracy          0.9103       15335
 macro avg         0.9098     0.9106     0.9101       15335
 weighted avg      0.9106     0.9103     0.9104       15335

Confusion Matrix:
[[6597  612]
 [ 763 7363]]
ROC-AUC Score: 0.9625522705281947
```

Fig 5.3 XGBoost classification Report

b) Random Forest

The Random Forest model showed strong performance with a testing accuracy of 90.71%. It identified both classes with balanced precision and recall scores around 0.90-0.91, indicating few misclassifications but slightly lower overall performance than XGBoost.

```
Random Forest Classification Report:
              precision    recall  f1-score   support

     0.0       0.8900      0.9157      0.9026       7209
     1.0       0.9232      0.8996      0.9112       8126

 accuracy          0.9071          15335
 macro avg       0.9066      0.9076      0.9069          15335
 weighted avg    0.9076      0.9071      0.9072          15335

Confusion Matrix:
[[6601  608]
 [ 816 7310]]
ROC-AUC Score: 0.9563894941261345
```

Fig 5.4 Random Forest classification Report

ROC Curve of each model

The ROC (Receiver Operating Characteristic) curve comparison illustrates the trade-off between the true positive rate (recall) and the false positive rate. The Area Under the Curve (AUC) is used to quantify each model's overall performance. Both models significantly outperform a random guess, but the higher AUC for XGBoost confirms its superior ability to distinguish between successful and failed startups.

a) XGBoost

With an **AUC of 0.963**, the XGBoost model achieved the highest score, indicating excellent discriminative power.

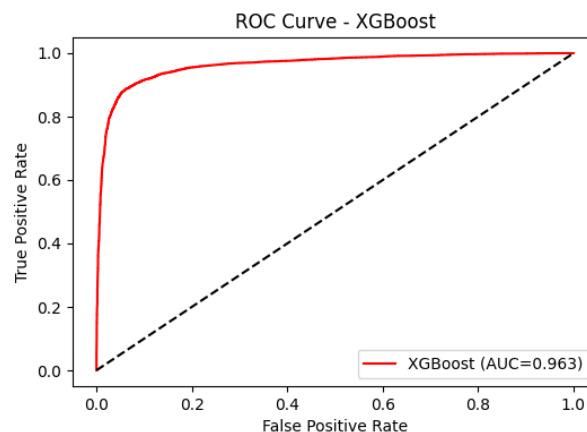


Fig 5.5 XGBoost ROC Curve

b) Random Forest

The Random Forest model followed closely with an **AUC of 0.956**, signifying near-perfect classification capability, though it was slightly outperformed by XGBoost.

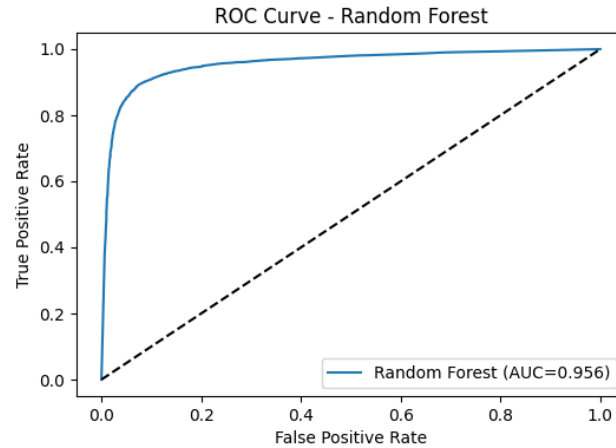


Fig 5.6 Random Forest classification Report

Feature Importance

An analysis of feature importance from both the XGBoost and Random Forest models provides critical insights into the key drivers of startup success or failure. Both models agree that a startup's financial health and operational history are the most significant factors.

a) XGBoost

The feature importance analysis from the selected XGBoost model reveals that financial, temporal, and categorical features all contribute to its predictions.

- The most influential feature by a significant margin is `funding_total_usd_log`, confirming that the total amount of capital raised is a primary determinant of a startup's fate.
- Other highly predictive features include `primary_category_analytics`, `company_age_years`, and various country codes such as `country_code_UNK` (unknown) and `country_code_RUS` (Russia). This indicates that both the industry sector and geographic location play important roles.

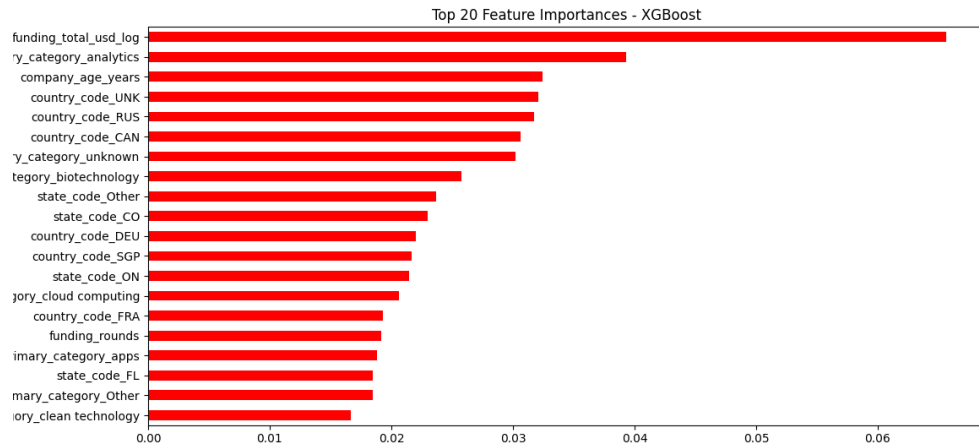


Fig 5.7 XGBoost Feature Importances

b) Random Forest

The Random Forest model also identifies financial and temporal metrics as the most critical predictors, concentrating most of its predictive power on a few key features.

- Similar to XGBoost, funding_total_usd_log is identified as the single most important predictor.
- The model also assigns high importance to company_age_years, funding_duration_years, and funding_rounds.
- Unlike XGBoost, which shows a more distributed importance, the Random Forest model indicates that these top four features are substantially more influential than all other categorical or geographic variables.

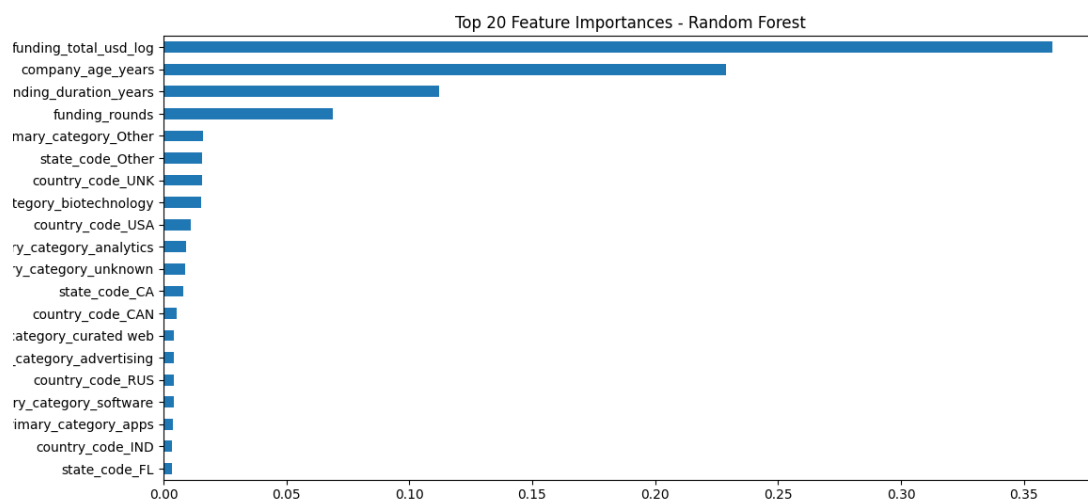


Fig 5.8 Random Forest Feature Importances

Result analysis and findings

```
Finovix Labs | Funding: $45,000,000 | Rounds: 8 | Category: FinTech
Country: GBR | State: LND | Founded: 2015-03-12 | First Funding: 2016-01-25
Last Funding: 2021-07-09 => Likely to Succeed
(Confidence: 0.94)
Medinova Care | Funding: $300,000 | Rounds: 1 | Category: HealthCare
Country: IND | State: KA | Founded: 2022-01-10 | First Funding: 2022-06-05
Last Funding: 2022-06-05 => At Risk of Failure
(Confidence: 0.90)
TrendCart | Funding: $8,000,000 | Rounds: 4 | Category: E-Commerce
Country: IND | State: MH | Founded: 2017-09-20 | First Funding: 2018-03-15
Last Funding: 2023-08-19 => Likely to Succeed
(Confidence: 0.97)
BlockMint Systems | Funding: $2,000,000 | Rounds: 2 | Category: Blockchain
Country: CAN | State: ON | Founded: 2019-06-22 | First Funding: 2020-02-14
Last Funding: 2021-09-09 => Likely to Succeed
(Confidence: 0.93)
```

Fig 5.9 final Test

A final test was performed using several hypothetical startup profiles to evaluate the real-world performance of the selected XGBoost model. The system is designed to provide a prediction of "Likely to Succeed" or "At Risk of Failure" along with a confidence level. The results showed that the model's predictions strongly align with the key financial indicators identified during training. For instance, startups with strong financial profiles like Finovix Labs (\$45M funding, 8 rounds) and TrendCart (\$8M funding, 4 rounds) were confidently predicted as 'Likely to Succeed' with high confidence scores of 0.94 and 0.97, respectively. Conversely, a startup with weak financial traction like Medinova Care (\$300k funding, 1 round) was correctly identified as 'At Risk of Failure' with 90% confidence. This consistency demonstrates the model's ability to reliably apply the learned patterns, where higher funding and more funding rounds strongly correlate with a positive outcome.

6. MODEL DEPLOYMENT

6.1 Model Implementation

The Start-up Failure Prediction System, named VentureGuard, is seamlessly integrated into a visually engaging and user-friendly web interface. This application is designed to make the risk assessment process simple, interactive, and insightful for investors and founders.

The application welcomes users with a clean, modern landing page that clearly states its purpose: "AI-Powered Predictions for Startup Success". This sets a professional and intuitive tone for interaction. Upon clicking "GET STARTED," users are directed to an input form where they are prompted to enter a startup's key details. These metrics include Funding Total, Funding Rounds, Primary Category, location data, and key dates like the Founded Date and Last Funding Date.

Once the data is submitted, the system processes it using the best-performing XGBoost machine learning model identified from the comparative study. The interface then presents the prediction in a clear, concise pop-up window. It displays the final outcome—either "Likely to Succeed" or "At Risk of Failure"—along with a Confidence Level and other calculated metrics like Company Age and Funding Duration.

The web application, developed using Python and Flask, ensures smooth navigation and provides real-time predictions. This integration successfully bridges the gap between complex machine learning analysis and practical, real-world application, making it a valuable tool for data-driven decision-making in the venture capital ecosystem.

6.2 User Interface

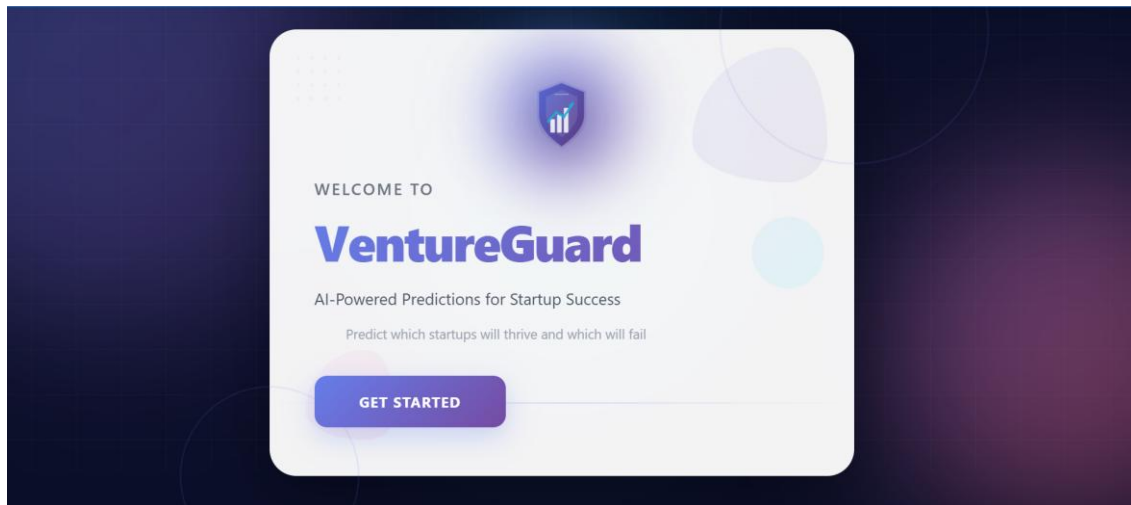


Fig 6.1 Welcome Page

Welcome Page: This is the initial landing page for the VentureGuard AI prediction tool. It features the application's name and its core promise: AI-Powered Predictions for Startup Success. The page's design is professional and tech-focused, aiming to establish trust in its data-driven mission to "Predict which startups will thrive and which will fail." The user's only action is to click the "GET STARTED" button to proceed.

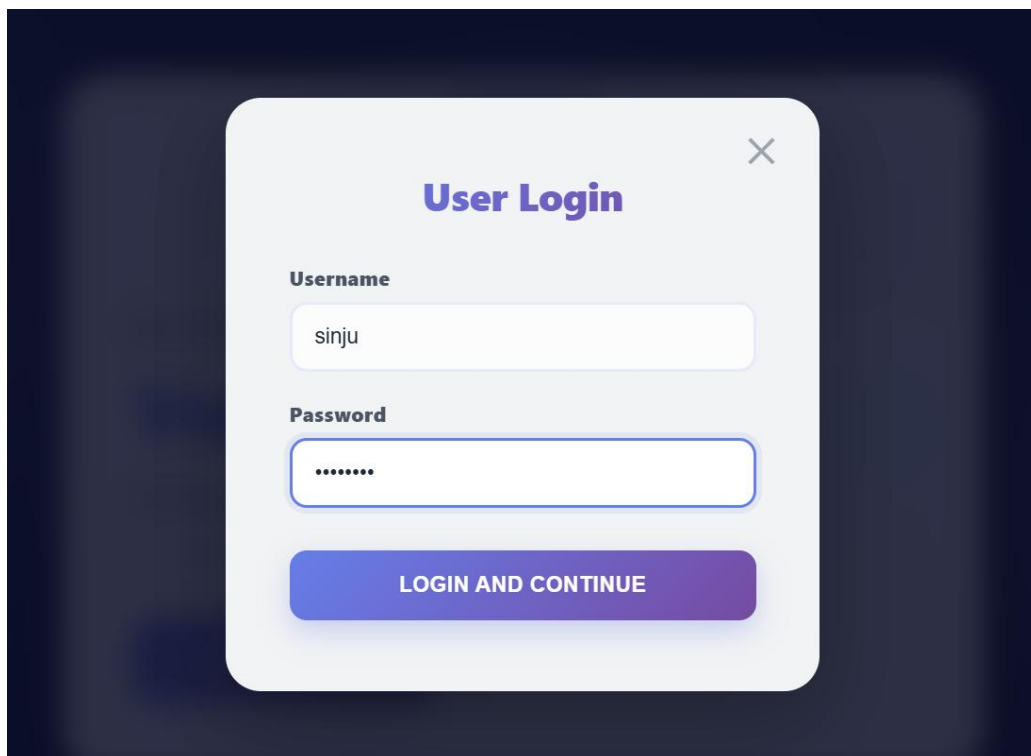
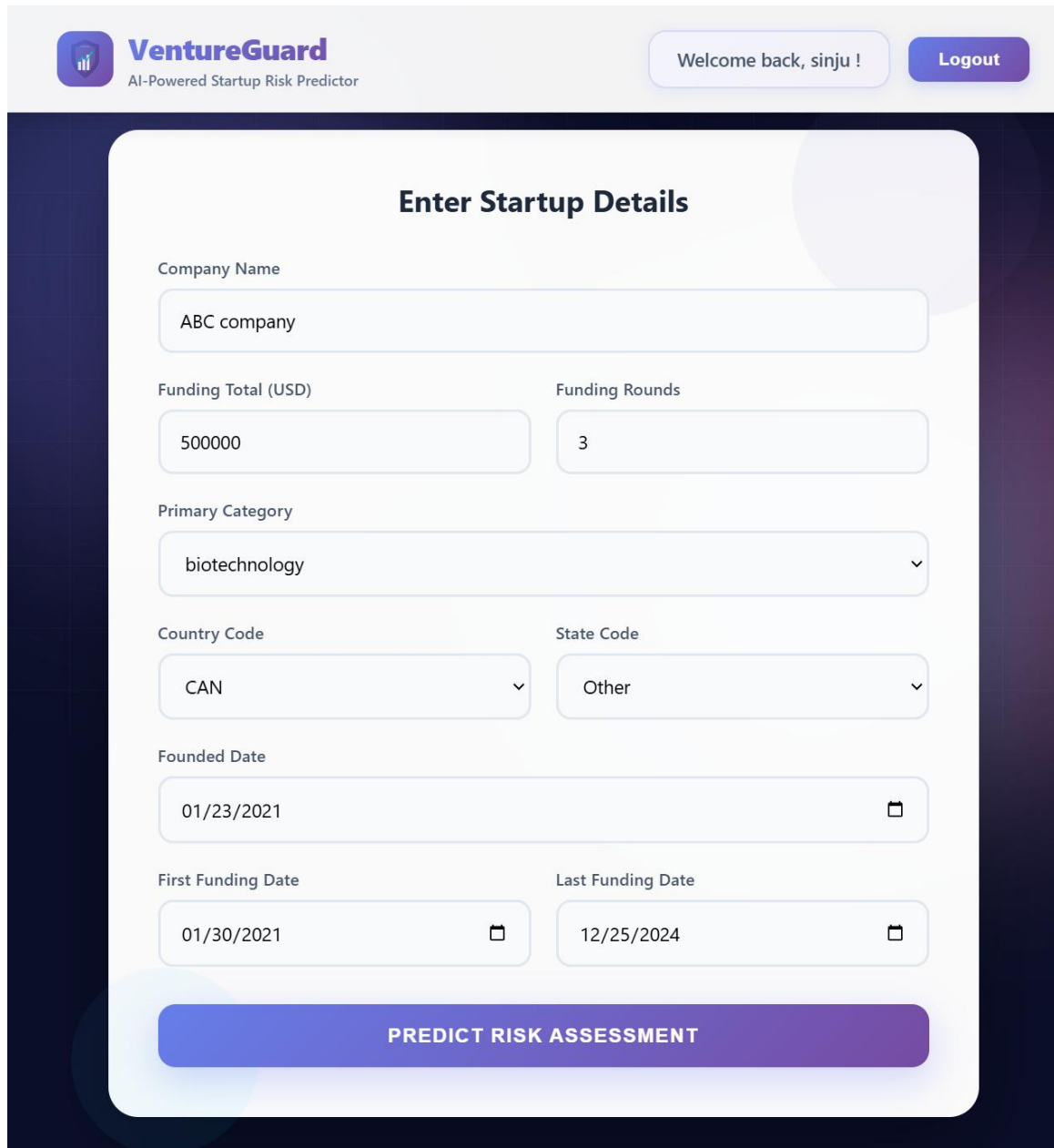


Fig 6.2 Login Page

User Login Page: This modal interface serves as the necessary authentication gateway for the application. It prompts the user to enter their Username and Password (e.g., sinju) to secure the application. After inputting credentials, the user clicks "LOGIN AND CONTINUE," successfully gaining access to the main prediction tool.



The screenshot displays the 'Enter Startup Details' form within the VentureGuard application. The header includes the VentureGuard logo (AI-Powered Startup Risk Predictor), a welcome message 'Welcome back, sinju !', and a 'Logout' button. The form itself is titled 'Enter Startup Details' and contains several input fields: 'Company Name' (text input with 'ABC company'), 'Funding Total (USD)' (text input with '500000'), 'Funding Rounds' (text input with '3'), 'Primary Category' (dropdown menu with 'biotechnology'), 'Country Code' (dropdown menu with 'CAN'), 'State Code' (dropdown menu with 'Other'), 'Founded Date' (date picker with '01/23/2021'), 'First Funding Date' (date picker with '01/30/2021'), and 'Last Funding Date' (date picker with '12/25/2024'). A large purple button at the bottom is labeled 'PREDICT RISK ASSESSMENT'.

Fig 6.3 Input Page

Startup Details Input Page: This is the main data entry form, titled "Enter Startup Details." The user is prompted to input specific, quantifiable data points crucial for the AI assessment, such as Funding Total, Funding Rounds, Primary Category, and Key Dates (Founded, First/Last

Funding). Once all required fields are populated, the user initiates the analysis by clicking the "PREDICT RISK ASSESSMENT" button.

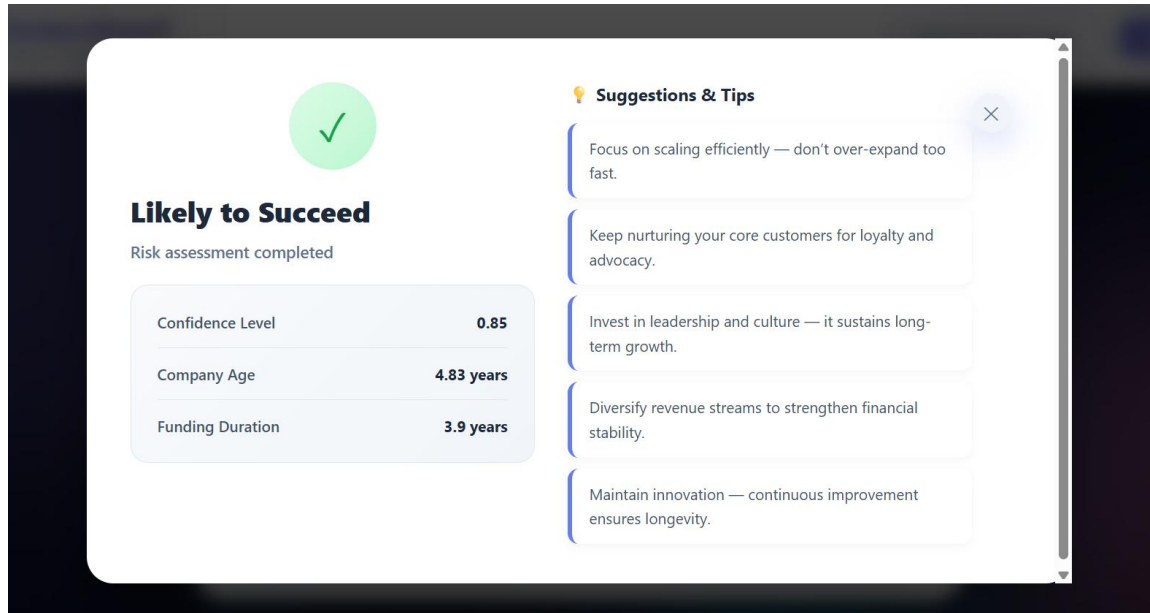


Fig 6.4 Result Page

Risk Assessment Results Page : This final page presents the AI model's output in a clear, actionable format. The primary finding is prominently displayed, showing the startup is "Likely to Succeed" with a high Confidence Level (0.85). The page also summarizes key input metrics like Company Age and Funding Duration, and provides practical "Suggestions & Tips" for sustaining long-term growth.

7. GIT HISTORY

The Git repository for the VentureGuard project includes all essential resources such as Python scripts, machine learning notebooks, the trained XGBoost and Random Forest models, the preprocessing pipeline, and the complete Crunchbase-based dataset used during development. It also contains the Flask backend files, HTML templates for the web interface, and the saved model along with the scaler, ensuring that future users can easily retrain, update, or extend the system. This well-organized and structured repository not only enhances project clarity and maintainability but also serves as a reliable reference point for debugging, evaluation, and future feature improvements.



```
commit 8b0705256faffd1f523b58f9b7d5fc3472030f04 (HEAD -> main, origin/main)
Author: Sinju-Mathews <sinjumathews0005@gmail.com>
Date: Thu Nov 20 23:43:09 2025 +0530

    login-logout

commit 1f7b722405354f556aa30bae71dd4df6646c269b
Author: Sinju-Mathews <sinjumathews0005@gmail.com>
Date: Thu Nov 20 23:06:29 2025 +0530

    Initial commit
```

Fig 7.1 Git Commit History

8. CONCLUSION

In conclusion, the VentureGuard project successfully fulfilled its objective of developing a reliable and practical machine learning system capable of predicting startup failure. Using a comprehensive Crunchbase-based dataset, the project followed a structured pipeline involving data cleaning, feature engineering, exploratory analysis, and model development. Two ensemble algorithms—Random Forest and XGBoost—were trained and compared to determine which offered the most robust performance for real-world use.

Among the models, XGBoost emerged as the stronger and more dependable predictor. With a testing accuracy of 91.03% and a recall of 90.61% for identifying failed startups, it demonstrated an excellent ability to capture high-risk cases—an essential requirement in risk analysis. Random Forest showed higher training accuracy but struggled to generalize, indicating overfitting. The superior stability and evaluation metrics of XGBoost make it the better choice for a decision-support system like VentureGuard.

Feature importance analysis revealed that financial factors—such as total funding, number of funding rounds, funding duration—and company age were the most influential predictors. These results align with established entrepreneurial patterns, highlighting how sustained funding, investor confidence, and longer operational history contribute to startup success. This interpretability adds practical value by helping users understand the drivers behind the predictions.

A key achievement of the project was deploying the trained XGBoost model into a Flask-based web application, making VentureGuard accessible and user-friendly. The application enables entrepreneurs, investors, incubators, and analysts to input startup characteristics and instantly receive clear risk evaluations, effectively bridging advanced machine learning with everyday decision-making.

Overall, VentureGuard demonstrates the potential of machine learning to strengthen risk assessment within the startup ecosystem. By combining a high-performing model, insightful feature analysis, and user-centric deployment, the project provides a meaningful tool that supports data-driven decisions, reduces uncertainty, and enhances strategic planning in the entrepreneurial landscape.

9. FUTURE SCOPE

The VentureGuard project offers a strong foundation, but several enhancements can further improve its accuracy, usability, and real-world relevance:

1. **Integration of Additional Features:** Incorporating more granular financial data, market indicators, social media sentiment, founder experience, and investor networks can enrich the model's predictive power.
2. **Transition to Deep Learning Models:** Advanced algorithms such as neural networks, LSTMs for time-series funding patterns, or graph neural networks for investment-founder relationships may capture complex patterns more effectively.
3. **Real-Time Data Updating:** Connecting the system to live APIs (e.g., Crunchbase, AngelList) would enable real-time monitoring and dynamic risk scoring as new funding rounds or company events occur.
4. **Interactive Dashboards:** Building a visual analytics dashboard with tools like Dash or Streamlit can help users explore feature importance, compare startups, and visualize risk scores more intuitively.
5. **Deployment as a Cloud-Based Service:** Hosting VentureGuard on platforms like AWS, Azure, or Google Cloud can allow secure multi-user access, scalability, and integration with external systems.
6. **Explainable AI (XAI) Enhancements:** Adding SHAP or LIME explanations can help users better understand how each feature contributes to the predicted risk score, improving trust and transparency.
7. **Mobile/Application Version:** Developing a mobile or cross-platform app can make the system more accessible for entrepreneurs, accelerators, and investors on the go.
8. **Industry-Specific Risk Models:** Creating separate models for sectors like fintech, healthtech, edtech, or e-commerce would allow domain-specific prediction with higher accuracy.

10. APPENDIX

10.1. Minimum Software Requirements

- Operating System: Windows 10 or above /macOS/Linux
- Python Version: Python 3.8 or above
- Browser: Google chrome/ Edge (for running UI)
- Server Execution: Flask application

10.2. Minimum Hardware Requirements

- Processor: Dual-core 1.6 GHz or higher
- RAM: Minimum 2GB or above
- Storage: 500 MB free space (for the application files and saved models)
- Display: Standard resolution (1366×768 or higher)

11. REFERENCE

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